SECP2523: DATABASE

SEM I 2024/2025

DATABASE DESIGN DESCRIPTIONS: LOGICAL DATABASE DESIGN

FOR KOPERASI KADA AUTOMATED SYSTEM

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1.0 Introduction

In the modern digital era, organizations like Koperasi KADA struggle with the inefficiencies and inaccuracies of paper-based systems for handling membership and loan applications. These traditional processes often lead to frequent errors and slow workflows. To resolve these challenges, introducing a computerized system has been recommended. By shifting to a digital platform, Koperasi KADA can enhance its operations, reduce mistakes, and manage data more effectively.

1.1. Overview about the Company

Lembaga Kemajuan Pertanian Kemubu (KADA) is a Malaysian organization dedicated to advancing the agricultural sector and supporting farming communities. While its primary mission is to promote agricultural development, KADA also operates a cooperative system that provides financial services to its members. These services include membership management, various loan offerings, saving accounts, and dividend distributions, all aimed at improving the financial well-being of its members and supporting their personal and professional growth.

The koperasi system offers diverse financial services, including Pinjaman Al-Bai, Pinjaman Al-Innah, and interest-free loans under the Pinjaman Qardhul Hasan scheme. These loan services are tailored to meet different financial needs, such as road tax payments, insurance expenses, and general financial support. Saving accounts managed by the cooperative also ensure members receive dividends, which are calculated with oversight from external auditors to maintain transparency and accuracy.

Despite its comprehensive offerings, KADA's current koperasi operations rely heavily on manual processes, which are time-consuming and prone to errors. Paper based systems for membership registration, loan applications, and financial record management pose challenges such as data inaccuracies and processing delays. However, KADA remains committed to enhancing its operations to provide more efficient and reliable services, ultimately contributing to its mission of fostering agricultural and financial growth in the community

1.2. The Existing System

Currently, Koperasi KADA depends on a manual and paper-based system in regard to managing membership and loan services provided. It involves writing everything on paper by hand regarding membership registration and loan applications. This is very labor-intensive and time-consuming hence, quite inefficient, especially with the increase in applications. This dependence on physical documentation comes along with issues such as reduced volumes of applications handled, delays in processing of applications and an increased chance of errors thereby dissatisfied the users.

One of the critical problems is the lack of an automated system for membership registration and loan applications. It hinders KADA Corporation from efficiently serving more applicants for both membership and loan. Approval statuses have to be conveyed by employees through telephone calls to the applicants, which is very time-consuming and adds to workload. In addition, recording information about applicants manually upon approval opens up their information to a high level of inaccuracy, such as misspelling or illegibility of handwriting.

The loans and statements are calculated and updated manually. This highly exposes the process to calculation eros and places heavy administrative work on the employees. Additionally, storing all members' and loans records on paper exposes data security to a high level of risk since the records are prone to theft, natural calamities and accidental losses due to negligence which could compromise information integrity.

1.3. The Proposed System

To address the challenges faced by KADA, we propose the digitalized of KADA system, which will modernize and streamline its core operations through five modules which are Membership Module, Terminate Membership Module, Review Module, Loan Application Module, Approve Application Module and Statement Report Production Module. Such a system will automate manual tasks to minimize errors and enhance overall efficiency for both members and employees of the Koperasi. The modules are as follows ;

1. Membership Module
   The module will let them automate membership registration and approval, applying online and tracking statuses of their applications in real-time.

2. Terminate Membership Module
   The module will ease the process of membership termination by allowing members to submit termination requests online. Members will have access to a real-time status tracker to monitor the progress of their termination requests.

3. Review Module
   The module is intended to help administrators in membership and loan reviews with the functionality to flag incomplete submissions and record detailed feedback.

4. Loan Application Module
   The module will manage the loan application process by featuring a user-friendly interface with the ability to submit loan requests, upload necessary documents and utilize a built-in loan calculator to estimate the amount of repayments.

5. Approve Application Module
   The module enables the ALK to evaluate applications reviewed by admin against predefined criteria for approval and rejection. This module will prompt ALK to make decision to reject or approve membership and loan application,

6. Statement Report Production Module
   The module produces comprehensive financial reports after the approval or rejection of membership and loan application. This module will also allow

admin to send the status of the applications and their statements to the applicants by email.

7. Financial tracking module

   The module is designed to manage transactions of applicants. It ensures that financial operations related to the system are accurately tracked and organized. This module also allows applicants to track their payment regarding their savings and loan.

These seven modules will work together to solve the existing operational challenges of the Koperasi and will ensure that the processing is more accurate, transparent and satisfactory for the members. It also focuses on reducing the dependence on paper records and manual labor. The KADA Automated System will go a long way in helping cost savings in the long term, while also enabling the organization to tackle increasing numbers of applications efficiency.

2.0 Global Conceptual Data Model (Global ERD)

The Global Conceptual Data Model, or Global ERD, is a combined diagram that merges smaller local data models into one big picture of the organization's data. It shows all the main parts of the system, their details, and how they are connected to each other. This model makes it easier to manage data, share information, and keep everything organized and running smoothly.

2.1. Global Conceptual Database Design (Conceptual ERD)

Figure 2.1 is the conceptual Entity-Relationship Diagram (ERD) representing the digitalized KADA system, detailing the relationships between users, memberships, loans and administrative processes. It includes entities such as User, Administrator, Applicant, Membership, MembershipEnd, Loan and associated modules like LoanType, Status and Feedback. Core relationships highlight processes like membership and loan applications, approval, reviews, terminations and repayments. The diagram also incorporates a tracking mechanism to monitor application updates and progress.

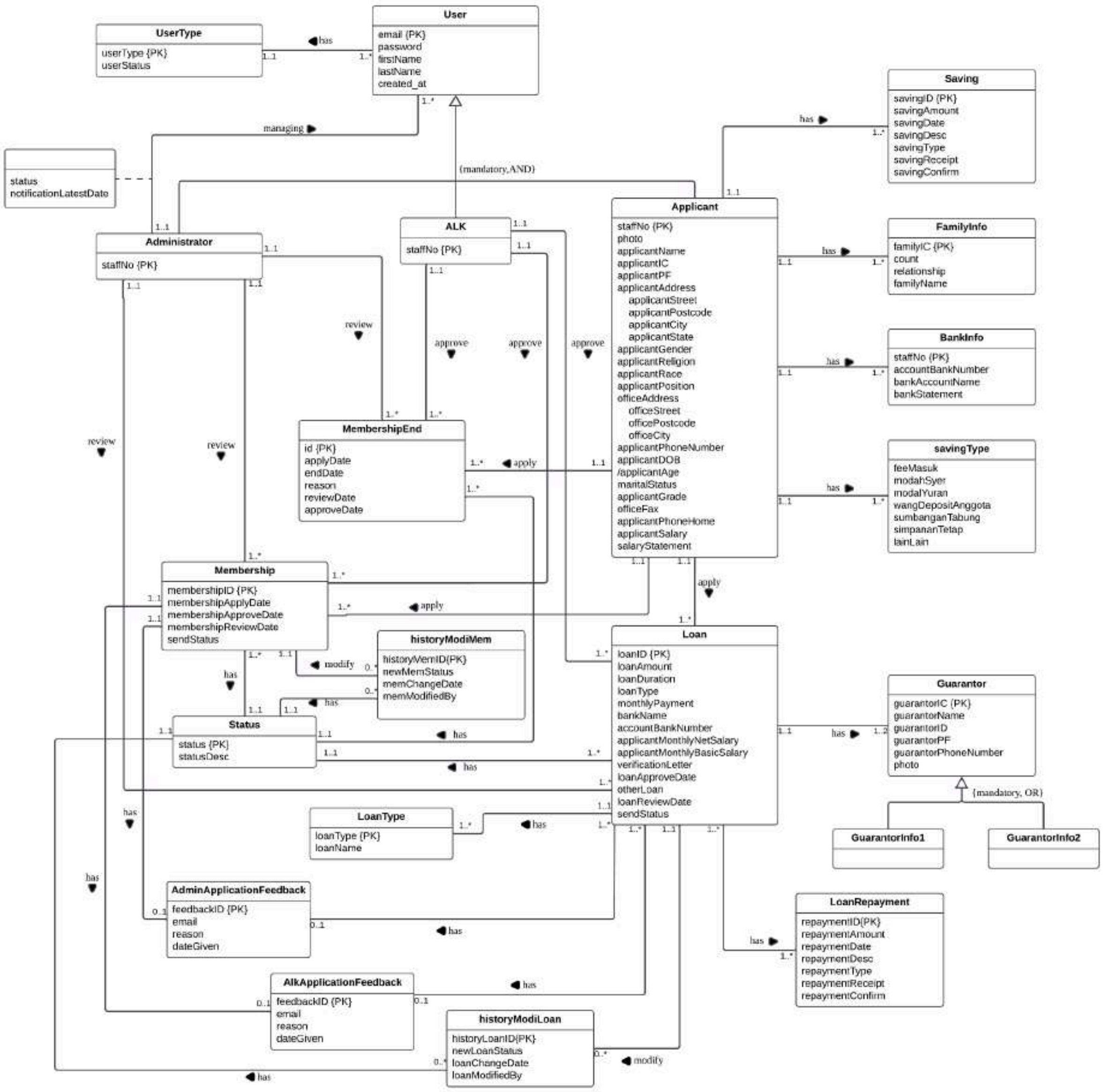


Figure 2.1 Conceptual ERD

2.2. Data Dictionary for global Conceptual ERD

A data dictionary includes important details about the entities, relationships, attributes, and attribute domains within a system. Entities represent objects or concepts, such as a customer or product, while relationships show how these entities are connected. Attributes define the properties of entities, like a staff's name or age. The attribute domain specifies the allowed values for each attribute, ensuring consistency and preventing errors. By clearly documenting these elements, a data dictionary helps maintain a well-structured, understandable system that ensures accurate and consistent data management.

2.2.1. Entity types

Entity Name	Description	Aliases	Occurrence
User	General term describing all the people who used the system.	End user	Each user will access the same system but different activities.
UserType	General term describing type of user who accesses the system	End user	Each userType will have different interface
Applicant	General term describing all users logged in as applicants to apply membership.	Membership Applicant	Each applicant will apply for membership after login to the system.
Administrator	General term describing the staff responsible for managing and reviewing membership and loan application	System Administrator	Each admin can manage multiple membership and loan applications
ALK	General term describing the authorized personnel responsible for approving membership and loan application	Approving Officer, Decision Maker	Each ALK is associated with membership and loan for approval
Membership	General term describing to the record of applicants who are members of the Koperasi and	Member Record, Applicant Membership	Each membership is only for one applicant
FamilyInfo	General term for the details of the family members associated with an applicant	Family Details, Household Information	Each applicant can provide multiple family member details
BankInfo	General term	Bank Account	Each applicant

	describing the bank account information provided by applicants for processing membership applications.	Details, Bank Information	must provide their bank information when applying for membership through the system.
Saving	General term describing the funds saved by members, managed through deposits and balance tracking used as collateral for loans.	Savings Record	Each saving record is associated with a single member of Koperasi KADA.
Loan	General term describing the financial agreement where an applicant borrows a specific amount of money to be repaid over a period with interest.	Loan Application, Credit Request	Each loan is uniquely tied to an applicant and involves submission, review and approval activities within the system.
Guarantor	General term describing a person who agrees to take responsibility for repaying a loan if the applicant has not paid for it.	Financial backer	Each loan application must include at least 2 guarantor details, depending on the loan type and policy requirements.
LoanType	General term describing the different categories or plans of loans offered	Loan Category	Each loan category is essential to each loan application, where the loan application belongs to a member of Koperasi KADA.
Managing	General term describing a notification update related to an approval process	Notification Log	Each notification is related to membership status and loan status for communication purpose

MembershipEnd	General term describing an application to end membership	Membership Termination	Submitted when a member decides to end membership
historyModiMem	General term describing the ability of ALK to change membership status	Membership Modification Log	Used to modify and track changes made to membership status
historyModiLoan	General term describing the ability of ALK to change loan status	Loan Modification Log	Used to modify and track changes made to loan status
Status	General term describing the current situation of the applications	Application Status	Each application have its own status to show real-time situation of each application
GuarantorInfo1	General term describing a person who agrees to take responsibility for repaying a loan if the applicant has not paid for it.	First financial backer	Each loan application must include First Guarantor
GuarantorInfo2	General term describing a person who agrees to take responsibility for repaying a loan if the applicant has not paid for it.	Second financial backer	Each loan application must include Second Guarantor
LoanRepayment	General term describing the amount of loan needed to be paid monthly	Repayment Amount	Each loan application have certain amount of loan repayment based on a specific calculation
ALKApplicationFeedback	General term describing the reason of each application rejection by ALK	Application Rejection Reason	Each application rejection have remarks

AdminApplication Feedback	General term describing the reason of each application rejection by Admin	Application Rejection Reason	Each application rejection have remarks
savingType	General term describing type of savings the company had	Savings Category	Each saving type is linked with the applicant membership

2.2.2. Relationship type

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
UserType	1..1	has	User	1..*
User	1..1 1..1 1..1	can be can be can be	ALK Admin Applicant	1..* 1..* 1..*
Applicant	1..1 1..1 1..1 1..1 1..1 1..1 1..1	Has Has Has Has Has Has Has	BankInfo Membership FamilyInfo Saving savingType Loan MembershipEnd	1..* 1..* 1..* 1..* 1..* 1..* 1..*
Admin	1..1 1..1 1..1 1..1	Review Review Managing Review	Membership Loan User MembershipEnd	1..* 1..* 1..* 1..*
ALK	1..1 1..1 1..1	Approve Approve Approve	Membership Loan MembershipEnd	1..* 1..* 1..*
MembershipEnd	1..*	Has	Status	1..1
Membership	1..* 1..1 1..1	Has Has Has	Status AdminApplication Feedback AlkApplicationFeedback	1..1 0..1 0..1

	1..1	Modify	historyModiMem	0..*
Loan	1..1 1..1 1..* 1..* 1..1 1..1 1..1	Has Has Has Has Has Has Modify	Guarantor LoanType LoanRepayment Status AdminApplication Feedback AlkApplicationFeedback historyModiLoan	1..2 1..* 1..* 1..1 0..1 0..1 0..*
Guarantor	1..1 1..1	Has Has	GurantorInfo1 GuarantorInfo2	0..1 0..1
historyModiMem	0..*	Has	Status	1..1
historyModiLoan	0..*	Has	Status	1..1

2.2.3. Attribute and Attribute Domain

Entity Name	Attributes	Description	Data Type and Length	Nulls	Multi-valued
Admin	staffNo	Uniquely identifies an administrator staff	INT (5)	No	No
AdminApplicationFeedback	feedbackID email reason dateGiven	Uniquely identifies feedback comment Email Feedback messages to send Date email was sent	INT (20) VARCHAR (30) LONGTEXT DATE	No No No No	No No No No
ALK	staffNo	Uniquely identifies a board of director staff	INT (5)	No	No
AlkApplicationFeedback	feedbackID	Uniquely identifies feedback comment	INT (20)	No	No

	email	Email	VARCHAR (30)	No	No
	reason	Feedback messages to send	LONGTEXT	No	No
	dateGiven	Date email was sent	DATE	No	No
Applicant	staffNo	Uniquely identifies applicant	INT (5)	No	No
	photo	Photo of applicant	LONGBLOB	No	No
	applicantName	Applicant full name	VARCHAR (255)	No	No
	applicantIC	Applicant IC number	VARCHAR (12)	No	No
	applicantPF	Applicant personal file number	BIGINT(10)	No	No
	applicantStreet	Applicant street home address	VARCHAR (200)	No	No
	applicantPostcode	Applicant home address postcode	INT (6)	No	No
	applicantCity	Applicant city for home address	VARCHAR (100)	No	No
	applicantState	Applicant state for home address	VARCHAR (100)	No	No
	applicantGender	Applicant gender	VARCHAR (10)	No	No
	applicantReligion	Applicant religion	VARCHAR (20)	No	No
	applicantRace	Applicant race	VARCHAR (20)	No	No
	applicantPosition	Applicant position	VARCHAR (50)	No No	No No

	officeStreet	Applicant office street	VARCHAR (255)	No	No
	officePostcode	Applicant office postcode	INT (6)	No	No
	officeCity	Applicant office city	VARCHAR (100)	No	No
	applicantPhoneNumber	Applicant phone number	VARCHAR (15)	No	Yes
	applicantDOB	Applicant date of birth	DATE	No	No
	applicantAge	Applicant age	INT (5)	No	No
	maritalStatus	Applicant marital status	VARCHAR (15)	No	No
	applicantGrade	Applicant grade	VARCHAR (10)	No	No
	officeFax	Applicant office fax number	VARCHAR (15)	No	No
	applicantPhoneHome	Applicant phone number	VARCHAR (15)	Yes	No
	applicantSalary	Applicant basic salary	DECIMAL(10,2)	No	No
	salaryStatement	Applicant salary statement	LONGBLOB	No	No
BankInfo	bankAccountName	Applicant bank account name	VARCHAR (40)	No	No
	bankStatement	Applicant bank statement	LONGBOB	No	No
	accountBankNumber	Applicant account bank number	BIGINT(25)	No	No
FamilyInfo	familyIC	Applicant's family IC number	VARCHAR (20)	No	No

	relationship	Applicant relationship	VARCHAR (25)	No	No
	familyName	Applicant's family name	VARCHAR (255)	No	No
	count	Count of family member	INT (15)	No	No
Guarantor	guarantorIC	Guarantor IC number	VARCHAR (12)	No	No
	guarantorName	Guarantor name	VARCHAR (100)	No	No
	guarantorID	Guarantor ID number	INT(5)	No	No
	guarantorPF	Guarantor PF number	INT(5)	No	No
	photo	Guarantor photo	LONGBOB	No	No
	guarantorPhoneNumber	Guarantor phone number	VARCHAR (11)	No	No
HistoriModiLoan	historyLoanID	Loan ID history	INT(5)	No	No
	newLoanStatus	Loan new status	INT(3)	Yes	No
	loanChangeDate	Loan change date	DATE	Yes	No
	loanModifiedBy	Loan modified by	INT(5)	Yes	No
HistoryModiMem	historyMemID	Membership ID history	INT(5)	No	No
	newMemStatus	Membership new status	INT(3)	Yes	No
	memChangeDate	Membership change date	DATE	Yes	No
	memModifiedBy	Membership modified by	INT(5)	Yes	No

Loan	loanID	Loan ID	INT(5)	No	No
	loanAmount	Loan amount	DECIMAL(10,2)	No	No
	loanDuration	Loan duration	INT(2)	No	No
	loanStatus	Loan status	INT(3)	No	No
	monthlyPayment	Monthly payment amount	DECIMAL(10,2)	No	No
	bankName	Bank name	VARCHAR(30)	No	No
	accountBankNumber	Account bank number	BIGINT(25)	No	No
	applicantNetSalary	Applicant net salary	DECIMAL(10,2)	No	No
	applicantBasicSalary	Applicant basic salary	DECIMAL(10,2)	No	No
	pengesahanMajikan	Applicant verified employer	LONGBLOB	No	No
	loanApproveDate	Loan approve date	DATE	Yes	No
	staffNo	Staff number	INT(5)	No	No
	loanType	Loan type	INT(7)	No	No
	otherLoan	Other loan type	VARCHAR(25)	Yes	No
	loanReviewDate	Loan review date	DATE	Yes	No
	loanApplyDate	Loan apply date	DATE	No	No
	sendStatus	Status for sent email	INT(3)	Yes	No
LoanRepayment	repaymentID repaymentAmount	Repayment ID Repayment amount	INT (5) DECIMAL(10,2)	No No	No No

	repaymentDate	Repayment date	DATE	No	No
	repaymentDesc	Repayment description	VARCHAR (200)	Yes	No
	repaymentType	Repayment type	VARCHAR (30)	No	No
	repaymentReceipt	Repayment receipt	VARCHAR (15)	Yes	No
	repaymentConfirm	Repayment confirmation	VARCHAR (15)	Yes	No
LoanType	loanType	Loan type	INT(7)	No	No
	loanName	Loan name	VARCHAR (30)	No	No
Managing	status	Status	VARCHAR (25)	Yes	No
	notificationLatestDate	Notification latest date	VARCHAR (25)	Yes	No
Membership	membershipStatus	Membership status	INT (3)	No	No
	membership ApplyDate	Membership apply date	DATE	No	No
	membership ApproveDate	Membership approve date	DATE	Yes	No
	membership ReviewDate	Membership review date	DATE	Yes	No
	sendStatus	Status for sent email	INT (3)	Yes	No
MembershipEnd	id	Uniquely identifies membership end application form	INT (5)	No	No
	applyDate	Membership apply date	DATE	No	No
	Membership end date	Membership end date	DATE	Yes	No

	reason	Reasons message ti end membership	VARCHAR (255)	No	No
	reviewDate	Membership end review date	DATE	Yes	No
	approveDate	Membership end approve date	DATE	Yes	No
	status	Membership end status	INT (3)	No	No
Saving	savingID	Uniquely identifies saving transaction	INT (5)	No	No
	savingAmount	Saving amount	DECIMAL(10,2)	No	No
	savingDate	Saving date	DATE	No	No
	savingDesc	Saving description	VARCHAR (200)	Yes	No
	savingType	Saving payment type	VARCHAR (30)	No	No
	savingReceipt	Saving receipt	VARCHAR (15)	Yes	No
	savingConfirm	Saving confirmation	VARCHAR (15)	Yes	No
SavingType	feeMasuk	Amount of fee masuk	DECIMAL(10,2)	No	No
	modahSyer	Amount of modah syer	DECIMAL(10,2)	No	No
	modalYuran	Amount of modal yuran	DECIMAL(10,2)	No	No
	wangDepositAnggota	Amount of wang deposit anggota	DECIMAL(10,2)	No	No
	sumbanganTabung	Amount of sumbangan tabung	DECIMAL(10,2)	No	No
	simpananTetap	Amount of simpanan tetap	DECIMAL(10,2)	No	No

	lainLain	Amount of other saving	DECIMAL(10,2)	No	No
Status	status	Status	INT (3)	No	No
	statusDesc	Status description	VARCHAR (25)	No	No
User	email	User email	VARCHAR (50)	No	No
	password	User password	VARCHAR (50)	No	No
	firstName	First name	VARCHAR (255)	No	No
	lastName	Last name	VARCHAR (255)	No	No
	created_at	created at	TIMESTAMP	No	No
UserType	userType	User type	INT (5)	No	No
	userStatus	User status	VARCHAR (10)	No	No

3.0 Logical Data Model (Logical ERD)

Logical Data Model (Logical ERD) represents the structure of an organization's data and its relationships in a high-level manner. It defines entities and their attributes, and the relationships between them to ensure data consistency and avoid data redundancy. This model acts as a link between the physical design of the database and real business operations.

3.1. Logical Database Design (Logical ERD)

Figure 2.2 is the logical Entity-Relationship Diagram (ERD) that outlines the detailed structure and relationships within the digitalized KADA system. The logical ERD expands on the conceptual ERD by further clarifying their relationships. Moreover, the logical ERD builds on the conceptual design by specifying how these entities interact in a structured way, making it more detailed and ready for database implementation.

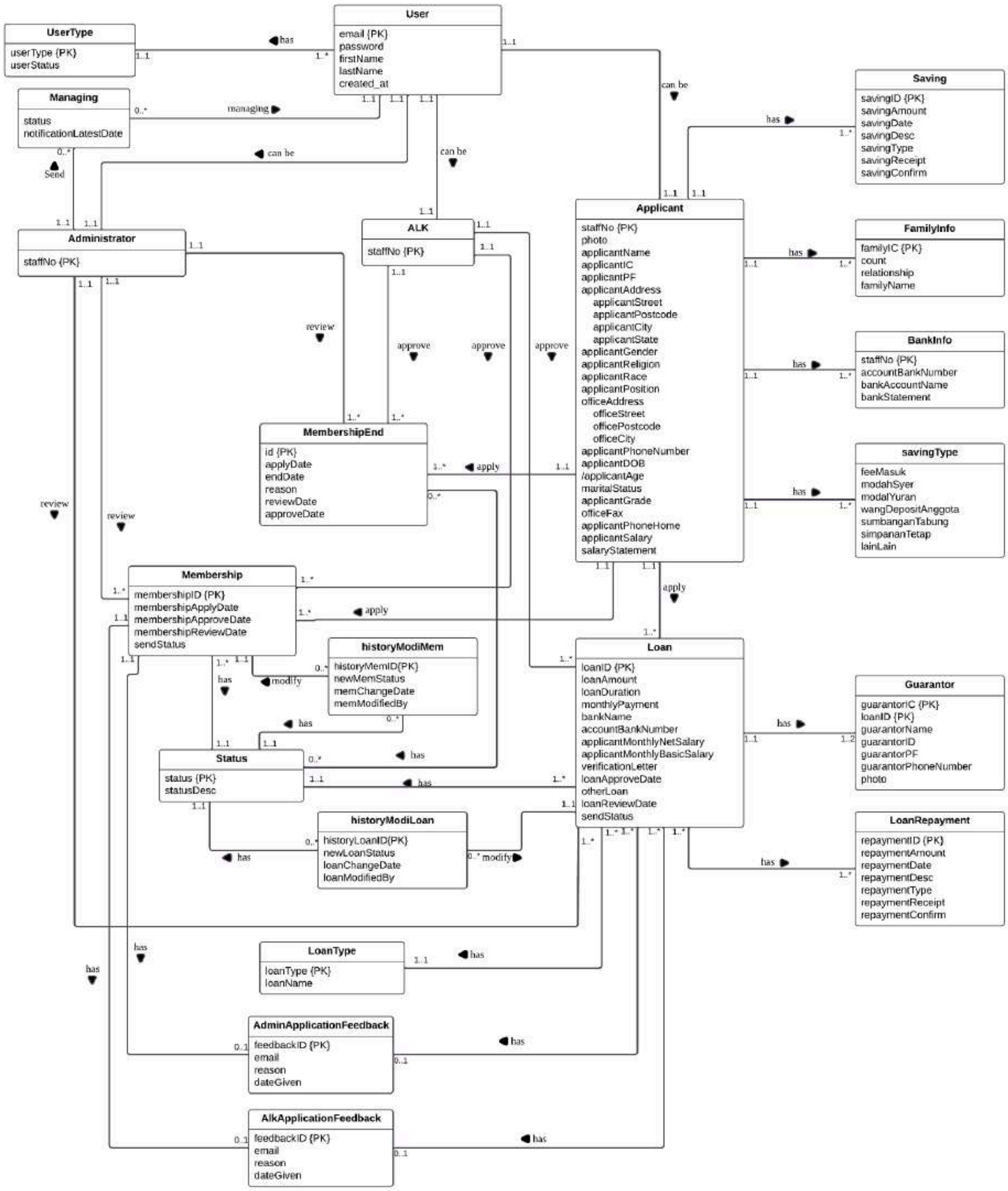


Figure 2.2 Logical ERD

3.2. Data dictionary for the Logical ERD

The Data Dictionary for the Logical ERD is a detailed guide that organizes information about entities, relationships, attributes, and attribute domains based on the normalized relations produced from the logical ERD. This dictionary serves as a reference to maintain clarity and consistency in the database structure.

3.2.1. Entity types

Entity Name	Description	Aliases	Occurrence
Admin	General term describing the staff responsible for managing and reviewing membership and loan application	System Administrator	Each admin can manage multiple membership and loan applications
AdminApplication Feedback	General term describing feedback provided by admin on membership or loan applications.	Admin Feedback, Application Feedback	Each feedback entry corresponds to one specific application.
ALK	General term describing the authorized personnel responsible for approving membership and loan application	Approving Officer, Decision Maker	Each ALK is associated with membership and loan for approval
AlkApplicationFeedback	General term describing feedback or decision provided by ALK on applications	ALK Feedback	Each feedback entry is linked to one application approved by ALK.
Applicant	General term describing all users logged in as applicants to apply membership.	Membership Applicant	Each applicant will apply for membership after login to the system.
BankInfo	General term describing the bank account information provided by applicants for processing membership applications.	Bank Account Details, Bank Information	Each applicant must provide their bank information when applying for membership through the system.
FamilyInfo	General term for the	Family Details,	Each applicant can

	details of the family members associated with an applicant	Household Information	provide multiple family member details
Guarantor	General term describing a person who agrees to take responsibility for repaying a loan if the applicant has not paid for it.	Financial backer	Each loan application must include at least 2 guarantor details, depending on the loan type and policy requirements.
HistoriModiLoan	General term describing changes made to loan details including update of the loan status.	Loan Modification History	Each loan modification is recorded and a loan application can have multiple modification records.
HistoryModiMem	General term describing tracking changes that are made to membership details including the update of the membership status.	Membership Modification History	Each membership modification is recorded and a membership can have multiple modification records.
Loan	General term describing the financial agreement where an applicant borrows a specific amount of money to be repaid over a period with interest.	Loan Application, Credit Request	Each loan is uniquely tied to an applicant and involves submission, review and approval activities within the system.
LoanRepayment	General term describing tracking repayment of the details for loans.	Loan Repayment Record	Each repayment record is associated with a specific loan.
LoanType	General term describing the different categories or plans of loans offered	Loan Category	Each loan category is essential to each loan application, where the loan application belongs to a

			member of Koperasi KADA.
Managing	General term describing process of admin manage users by sending notifications about their application	User Management	Each administrator can manage multiple users by sending notifications and each user can get notifications by only one admin.
Membership	General term describing to the record of applicants who are members of the Koperasi and	Member Record, Applicant Membership	Each membership is only for one applicant
MembershipEnd	General term describing details of membership termination including reason, date of application, approval and review.	Membership Termination	Each member can submit one termination request, which is then used to review and approve.
Saving	General term represents transaction records of saving.	Member Savings record.	Each saving record is linked to a specific member and store the details of their savings transaction.
SavingType	General term describing the funds saved by members, managed through deposits and balance tracking used as collateral for loans.	Savings Record	Each saving record is associated with a single member of Koperasi KADA.
Status	General term describing current stage or progress of a membership or loan application	Application Status, Membership Status, Loan Status	Each status is associated with one membership or loan application.
User	General term	End user	Each user will

	describing all the people who used the system.		access the same system but different activities.
UserType	General term describing user in the system, such as staff, admin or Ahli Lembaga Koperasi (ALK)	User category	Each user belongs to one user type and a user type can be assigned to multiple users.

3.2.2. Relationship type

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
User	1..* 1..1 1..1 1..1	has can be can be can be	UserType ALK Administartor Applicant	1..1 1..1 1..1 1..1
Applicant	1..1 1..1 1..1 1..1 1..1 1..1 1..1	has has has has apply apply apply	BankInfo FamilyInfo Saving savingType Loan Membership MembershipEnd	1..* 1..* 1..* 1..* 1..* 1..* 1..*
Administrator	1..1 1..1 1..1 1..1	send review review review	Managing Membership Loan MembershipEnd	0..* 1..* 1..* 1..*
ALK	1..1 1..1 1..1	approve approve approve	Membership Loan MembershipEnd	1..* 1..* 1..*
MembershipEnd	0..*	has	Status	0..*
Membership	1..* 1..1 1..1	has has has	Status AdminApplication Feedback AlkApplicationFeedback	1..1 0..1 0..1
Loan	1..1 1..* 1..* 1..*	has has has has	Guarantor LoanType LoanRepayment Status	1..2 1..1 1..* 1..1

	1..* 1..*	has has	AdminApplication Feedback AlkApplicationFeedback	0..1 0..1
historyModiLoan	0..* 0..*	modify has	Loan Status	1..1 1..1
historyModiMem	0..* 0..*	modify has	Membership Status	1..1 1..1
Managing	0..*	managing	User	1..1

3.2.3. Attribute and Attribute Domain

Entity Name	Attributes	Description	Data Type and Length	Nulls	Multi-valued
Admin	staffNo	Uniquely identified for admin staff number.	INT (5)	No	No
AdminApplicationFeedback	feedbackID email reason dateGiven	Uniquely identifies feedback comments. Email Feedback messages to send Date email was sent	INT (20) VARCHAR (30) LONGTEXT DATE	No No No No	No No No No
ALK	staffNo	Uniquely identified for ALK staff numbers.	INT (5)	No	No
AlkApplicationFeedback	feedbackID email reason	Uniquely identifies feedback comments. Email Feedback messages to send	INT (20) VARCHAR (30) LONGTEXT	No No No	No No No

	dateGiven	Date email was sent	DATE	No	No
Applicant	staffNo	Uniquely identifies applicant	INT (5)	No	No
	photo	Photo of applicant	LONGBOB	No	No
	applicantName	Applicant's full name	VARCHAR(255)	No	No
	applicantIC	Applicant IC Number	VARCHAR (12)	No	No
	applicantPF	Applicant personal file number	BIGINT(10)	No	No
	applicantStreet	Applicant street address	VARCHAR(200)	No	No
	applicantPostcode	Applicant home address postcode	INT (6)	No	No
	applicantCity	Applicant city for home address	VARCHAR(100)	No	No
	applicantState	Applicant state for home address	VARCHAR(100)	No	No
	applicantGender	Applicant gender	VARCHAR (10)	No	No
	applicantReligion	Applicant Religion	VARCHAR (20)	No	No
	applicantRace	Applicant Race	VARCHAR (20)	No	No
	applicantPosition	Applicant Position	VARCHAR (50)	No	No
	officeStreet	Applicant office street	VARCHAR(255)	No	No
	officePostcode	Applicant office postcode	INT (6)	No	No
	officeCity	Applicant office city	VARCHAR(100)	No	No
	applicantPhoneNumber	Applicant phone number	VARCHAR(15)	No	Yes
	applicantDOB	Applicant date of birth	DATE	No	No

	applicantAge	Applicant Age	INT (5)	No	No
	maritalStatus	Applicant Marital status	VARCHAR(15)	No	No
	applicantGrade	Applicant Grade	VARCHAR(10)	No	No
	officeFax	Applicant office fax number	VARCHAR(15)	No	No
	applicantPhoneHome	Applicant home phone number	VARCHAR(15)	Yes	No
	applicantSalary	Applicant Salary	DECIMAL(10,2)	Yes	No
	salaryStatement	Applicant Salary statement	LONGBOB	No	No
BankInfo	staffNo	Applicant staff number	INT (5)	No	No
	bankAccountName	Applicant bank account number	VARCHAR (40)	No	No
	bankStatement	Applicant bank statement	LONGBOB	No	No
	accountBankNumber	Applicant account bank name	BIGINT(25)	No	No
FamilyInfo	familyIC	Applicant's family IC number	VARCHAR(20)	No	No
	relationship	Applicant relationship	VARCHAR(25)	No	No
	familyName	Applicant's family name	VARCHAR(255)	No	No
	staffNo	Applicant staff number	INT (5)	No	No
	count	Count of family member	INT (15)	No	No
Guarantor	guarantorIC	Guarantor IC number	VARCHAR (12)	No	No

	guarantorName	Guarantor Name	VARCHAR(100)	No	No
	guarantorID	Guarantor ID number	INT(5)	No	No
	guarantorPF	Guarantor personal file number	INT(5)	No	No
	photo	Guarantor photo	LONGBOB	No	No
	loanID	loan identification number	INT (5)	No	No
	guarantorPhoneNumber	Guarantor Phone number	VARCHAR (11)	No	No
HistoriModiLoan	historyLoanID	Loan ID history	INT(5)	No	No
	newLoanStatus	Loan new status	INT(3)	Yes	No
	loanChangeDate	Loan change date	DATE	Yes	No
	loanModifiedBy	Loan modified by	INT(5)	Yes	No
	loanID	loan identification number	INT(5)	No	No
HistoryModiMem	historyMemID	Membership ID history	INT(5)	No	No
	newMemStatus	Membership new status	INT(3)	Yes	No
	memChangeDate	Membership change date	DATE	Yes	No
	memModifiedBy	Membership modified by	INT(5)	Yes	No
	membershipID	Membership identification number	INT(5)	No	No
Loan	loanID	loan identification number	INT(5)	No	No
	loanAmount	Loan Amount	DECIMAL(10,2)	No	No
	loanDuration	Loan Duration	INT(2)	No	No

	loanStatus	Loan status	INT(3)	No	No
	monthlyPayment	Monthly payment of loan	DECIMAL(10,2)	No	No
	bankName	Bank name	VARCHAR(30)	No	No
	accountBankNumber	Account bank number	BIGINT(25)	No	No
	applicantNetSalary	Applicant net salary	DECIMAL(10,2)	No	No
	applicantBasicSalary	Applicant basic salary	DECIMAL(10,2)	No	No
	pengesahanMajikan	Applicant verified employer	LONGBOB	No	No
	loanApproveDate	Loan approve date	DATE	Yes	No
	staffNo	Staff number	INT(5)	No	No
	loanType	Loan type	INT(7)	No	No
	otherLoan	Other loan type	VARCHAR(25)	Yes	No
	loanReviewDate	Loan review date	DATE	Yes	No
	loanApplyDate	Loan apply date	DATE	No	No
	adminStaffNo	Admin staff number	INT(5)	Yes	No
	alkStaffNo	ALK staff number	INT(5)	Yes	No
	sendStatus	Status for send email	INT(3)	Yes	No
LoanRepayment	repaymentID	Repayment ID	INT (5)	No	No
	repaymentAmount	Repayment amount	DECIMAL(10,2)	No	No
	repaymentDate	Repayment date	DATE	No	No
	repaymentDesc	Repayment description	VARCHAR(200)	Yes	No
	repaymentType	Repayment type	VARCHAR(30)	No	No

	repaymentReceipt repaymentConfirm loanID	Repayment receipt Repayment confirmation loan identification number	VARCHAR(15) VARCHAR(15) INT (5)	Yes Yes No	No No No
LoanType	loanType loanName	Loan type Loan name	INT(7) VARCHAR(30)	No No	No No
Managing	status notificationLatestDate	Status Notification latest date	VARCHAR(25) VARCHAR(25)	Yes Yes	No No
Membership	membershipStatus membershipApplyDate membershipApproveDate membershipReviewDate staffNo alkStaffNo adminStaffNo sendStatus	Membership status Membership apply date Membership approve date Membership review date staff number ALK staff number admin staff number Status for sent email	INT (3) DATE DATE DATE INT (5) INT (5) NT (5) INT (3)	No No Yes Yes No Yes Yes Yes	No No No No No No No No
Membership End	id applyDate endDate reason	Uniquely identifies membership end application form Membership apply date Membership end date Reasons message to end membership	INT (5) DATE DATE VARCHAR(255)	No No Yes No	No No No No

	reviewDate	reviewDate	DATE	Yes	No
	approveDate	approveDate	DATE	Yes	No
	staffNo	staff number	INT (5)	No	No
	adminStaffNo	admin staff number	INT (5)	Yes	No
	alkStaffNo	ALK staff number	INT (5)	Yes	No
	status	status	INT (3)	No	No
Saving	savingID	Uniquely identifies saving transaction	INT (5)	No	No
	savingAmount	Saving amount	DECIMAL(10,2)	No	No
	savingDate	Saving date	DATE	No	No
	savingDesc	Saving description	VARCHAR(200)	Yes	No
	savingType	Saving payment type	VARCHAR (30)	No	No
	savingReceipt	Saving receipt	VARCHAR (15)	Yes	No
	savingConfirm	Saving confirmation	VARCHAR (15)	Yes	No
	staffNo	staff number	INT (5)	No	No
SavingType	feeMasuk	Amount of fee masuk	DECIMAL(10,2)	No	No
	modahSyer	Amount of modah syer	DECIMAL(10,2)	No	No
	modalYuran	Amount of modal yuran	DECIMAL(10,2)	No	No
	wangDepositAnggota	Amount of wang deposit anggota	DECIMAL(10,2)	No	No
	sumbanganTabung	Amount of sumbangan tabung	DECIMAL(10,2)	No	No
	simpananTetap	Amount of simpanan tetap	DECIMAL(10,2)	No	No

	lainLain staffNo	Amount of other saving staff number	DECIMAL(10,2) INT (5)	No No	No No
Status	status statusDesc	Status Status description	INT (3) VARCHAR (25)	No No	No No
User	email password firstName lastName created_at	Email Password Applicant first name Applicant last name Date of account created	VARCHAR(50) VARCHAR(50) VARCHAR(255) VARCHAR(255) TIMESTAMP	No No No No No	No No No No No
UserType	userType userStatus	User type User status	INT (5) VARCHAR(10)	No No	No No

3.3. Relational Database Schema(s)

The relational database schema is designed to organize data into tables with each table representing an entity and its details stored in rows and columns. Tables are connected using primary keys and foreign keys to ensure data is related and consistent. This structure helps store, manage and retrieve data efficiently to make it useful for systems that handle connected information like memberships and loans. Below is the relational database schema related to the logical ERD for Koperasi KADA Automated System:

1. Administrator (staffNo)
   Primary Key: staffNo

2. AdminApplicationFeedback (feedbackID,email, reason, dateGiven)

   Primary Key: feedbackID

3. ALK (staffNo)

   Primary Key: staffNo

4. AlkApplicationFeedback (feedbackID,email, reason, dateGiven)

   Primary Key: feedbackID

5. Applicant (staffNo,photo, applicantName, applicantIC, applicantPF, applicantStreet, applicantPostcode, applicantCity, applicantState, applicantGender, applicantReligion, applicantRace, applicantPosition, officeStreet, officePostcode, officeCity, applicantPhoneNumber, applicantDOB, applicantAge, maritalStatus, applicantGrade, officeFax, applicantPhoneHome, applicantSalary, salaryStatement)

   Primary Key: staffNo

6. BankInfo (staffNo, accountBankNumber, bankAccountName, bankStatement)

   Primary Key: staffNo
   Foreign Key: staffNo References Applicant (staffNo)

7. FamilyInfo (familyIC, count, relationship, familyName, staffNo)

   Primary Key: familyIC
   Foreign Key: staffNo References Applicant (staffNo)

8. Guarantor (guarantorIC, loanID, guarantorName, guarantorID, guarantorPF, guarantorPhoneNumber, photo)

   Primary Key: guarantorIC, loanID
   Foreign Key: loanID References Loan (loanID)

9. HistoryModiLoan (historyLoanID, newLoanStatus, loanChangeDate, loanModifiedBy, loanID)

   Primary Key: historyLoanID
   Foreign Key: loanID References Loan (loanID)

10. HistoryModiMem (historyMemID, newMemStatus, memChangeDate, memModifiedBy, membershipID)

    Primary Key: historyMemID
    Foreign Key: membershipID References Membership (membershipID)

11. Loan (loanID, loanAmount, loanDuration, monthlyPayment, bankName, accountBankNumber, applicantMonthlyNetSalary, applicantMonthlyBasicSalary, verificationLetter, loanApproveDate, otherLoan, loanReviewDate, sendStatus, status, staffNo, adminStaffNo, alkStaffNo, loanType, adminFeedbackID, alkFeedbackID, repaymentID)

    Primary Key: loanID
    Foreign Key: status References Status (status)
    staffNo References Applicant (staffNo)
    adminStaffNo References Admin (staffNo)
    alkStaffNo References ALK (staffNo)
    adminFeedbackID References Admin (feedbackID)
    alkFeedbackID References ALK (feedbackID)
    repaymentID References LoanRepayment (repaymentID)

12. LoanRepayment (repaymentID, repaymentAmount, repaymentDate, repaymentDesc, repaymentType, repaymentReceipt, repaymentConfirm, loanID)

Primary Key: repaymentID

Foreign Key: loanID References Loan (loanID)

13. LoanType (<u>loanType</u>, loanName)

    Primary Key: loanType

14. Managing (status, notificationLatestDate, email, staffNo)

    Foreign Key: email References User (email)
    staffNo References Admin (staffNo)

15. Membership (<u>membershipID</u>, membershipApplyDate, membershipApproveDate, membershipReviewDate, sendStatus, status, staffNo, adminStaffNo, alkStaffNo, adminFeedbackID, alkFeedbackID)

    Foreign Key: status References Status (status)
    staffNo References Applicant (staffNo)
    adminStaffNo References Admin (staffNo)
    alkStaffNo References ALK (staffNo)
    adminFeedbackID References Admin (feedbackID)
    alkFeedbackID References ALK (feedbackID)

16. MembershipEnd (<u>id</u>, applyDate, endDate, reason, reviewDate, approveDate, status, staffNo, adminStaffNo, alkStaffNo)

    Foreign Key: status References Status (status)
    staffNo References Applicant (staffNo)
    adminStaffNo References Admin (staffNo)
    alkStaffNo References ALK (staffNo)

17. Saving (savingID, savingAmount, savingDate, savingDesc, savingType, savingReceipt, savingConfirm, staffNo)

    Primary Key: savingID
    Foreign Key: staffNo References Applicant (staffNo)

18. SavingType (feeMasuk, modahSyer, modalYuran, wangDepositAnggota, sumbanganTabung, simpananTetap, lainLain, staffNo)

    Foreign Key: staffNo References Applicant (staffNo)

19. Status (status, statusDesc)

    Foreign Key: status

20. User (email, password, firstName, lastName, created_at, userType)

    Primary Key: email
    Foreign Key: userType References UserType (UserType)

21. UserType (userType, userStatus)

    Primary Key: userType

3.4. Normalization

This section represents the process of organizing a database to ensure that data is stored logically and efficiently. It shows how data can be structured to eliminate redundancy, prevent anomalies and maintain consistency By following specific norms from 1NF to 2NF, this section demonstrates how relationships between data are clarified and dependencies are properly managed resulting in a database that is easier to maintain and query.

1NF

Since there is no repeating group and all tables have primary keys defined, all the tables satisfy 1NF

2NF

Partial dependencies :

All attributes are fully dependent on the respective primary key. All the tables do not have composite keys, thus the tables already satisfy 2NF

3NF

In all tables :

All non-key attributes directly depend on the primary key. No transitive dependencies detected. Thus, the tables satisfy 3NF. Below is the finalize tables for the database

1. Administrator (staffNo)
2. AdminApplicationFeedback (feedbackID,email, reason, dateGiven)
3. ALK (staffNo)
4. AlkApplicationFeedback (feedbackID,email, reason, dateGiven)

5. Applicant (staffNo,photo, applicantName, applicantIC, applicantPF, applicantStreet, applicantPostcode, applicantCity, applicantState, applicantGender, applicantReligion, applicantRace, applicantPosition, officeStreet, officePostcode, officeCity, applicantPhoneNumber, applicantDOB, applicantAge, maritalStatus, applicantGrade, officeFax, applicantPhoneHome, applicantSalary, salaryStatement)
6. BankInfo (staffNo, accountBankNumber, bankAccountName, bankStatement)
7. FamilyInfo (familyIC, count, relationship, familyName, staffNo)
8. Guarantor (guarantorIC, loanID, guarantorName, guarantorID, guarantorPF, guarantorPhoneNumber, photo)

9. HistoryModiLoan (historyLoanID, newLoanStatus, loanChangeDate, loanModifiedBy, loanID)
10. HistoryModiMem (historyMemID, newMemStatus, memChangeDate, memModifiedBy, membershipID)
11. Loan (loanID, loanAmount, loanDuration, monthlyPayment, bankName, accountBankNumber, applicantMonthlyNetSalary, applicantMonthlyBasicSalary, verificationLetter, loanApproveDate, otherLoan, loanReviewDate, sendStatus, status, staffNo, adminStaffNo, alkStaffNo, loanType, adminFeedbackID, alkFeedbackID, repaymentID)
12. LoanRepayment (repaymentID, repaymentAmount, repaymentDate, repaymentDesc, repaymentType, repaymentReceipt, repaymentConfirm, loanID)
13. LoanType (loanType, loanName)
14. Managing (status, notificationLatestDate, email, staffNo)
15. Membership (membershipID, membershipApplyDate, membershipApproveDate, membershipReviewDate, sendStatus, status, staffNo, adminStaffNo, alkStaffNo, adminFeedbackID, alkFeedbackID)
16. MembershipEnd (id, applyDate, endDate, reason, reviewDate, approveDate, status, staffNo, adminStaffNo, alkStaffNo)
17. Saving (savingID, savingAmount, savingDate, savingDesc, savingType, savingReceipt, savingConfirm, staffNo)
18. SavingType (feeMasuk, modahSyer, modalYuran, wangDepositAnggota, sumbanganTabung, simpananTetap, lainLain, staffNo)
19. Status (status, statusDesc)
20. User (email, password, firstName, lastName, created_at, userType)
21. UserType (userType, userStatus)

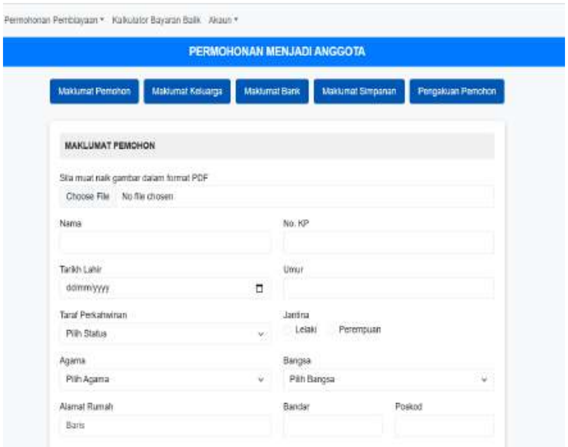
4.0 Interface Design and SQL implementation

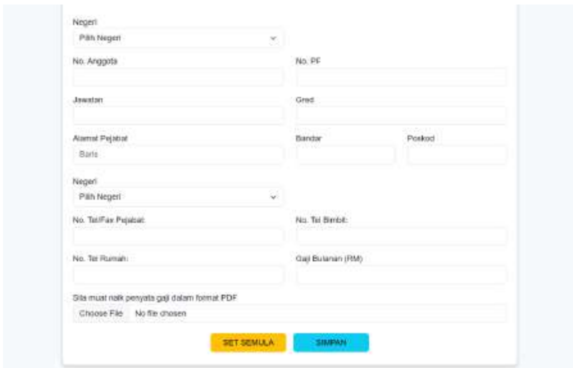
This section explains how the user interface connects with the database to make everything work smoothly. It shows how different parts of the interface are linked to specific SQL commands that interact with the database. This section also covers the SQL commands used to create the database structure and manage the data.

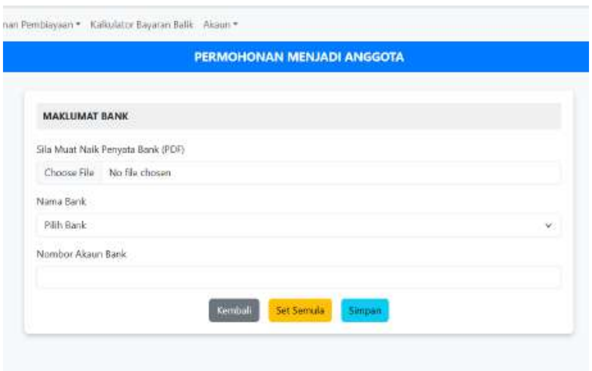
4.1. Interface Design

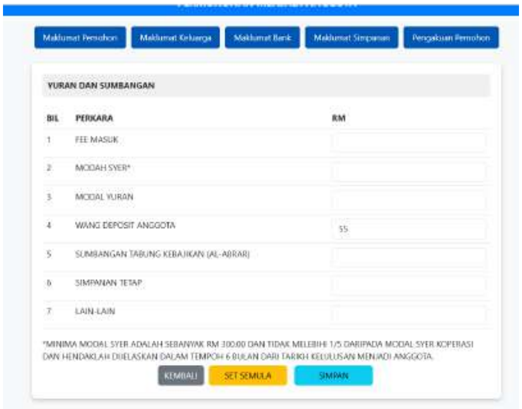
This section shows the interface design of the system is simple and user-friendly, allowing users to navigate easily through its features. The design ensures that users can complete tasks efficiently without unnecessary complexity.

4.1.1. Membership Application Module

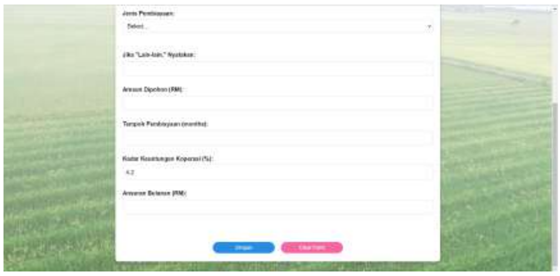




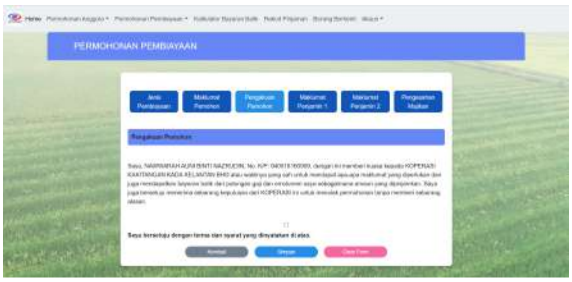


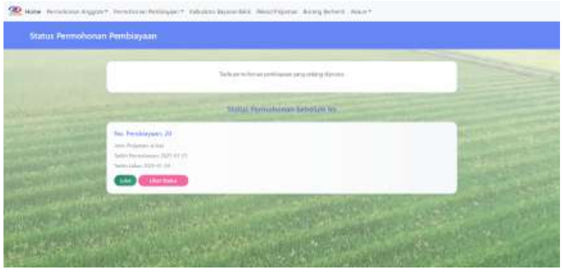


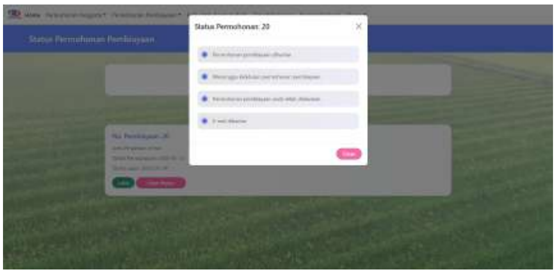
4.1.2. Loan Application Module

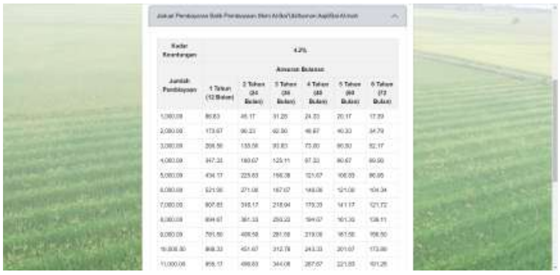




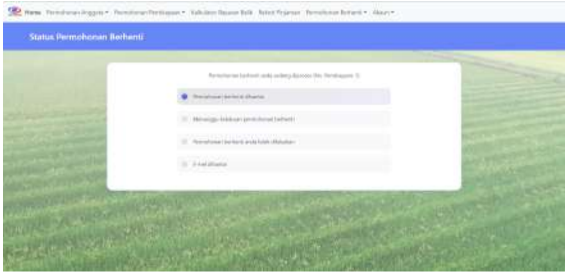







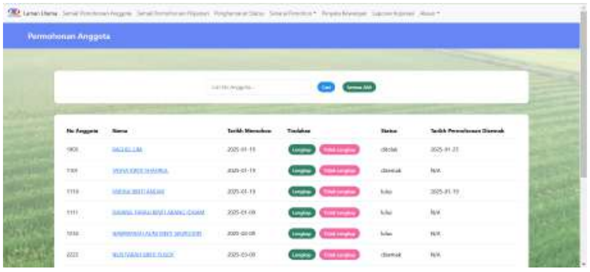
4.1.3. Terminate Module

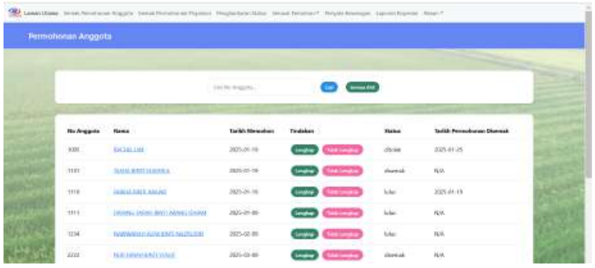


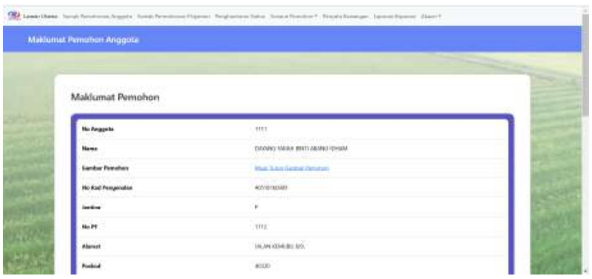


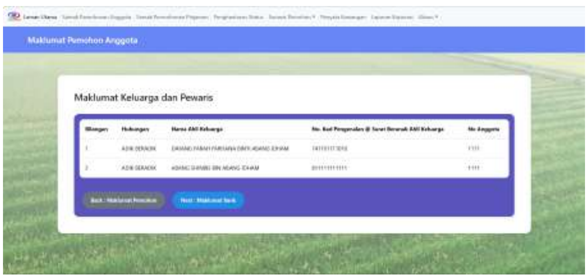
4.1.4. Review Module

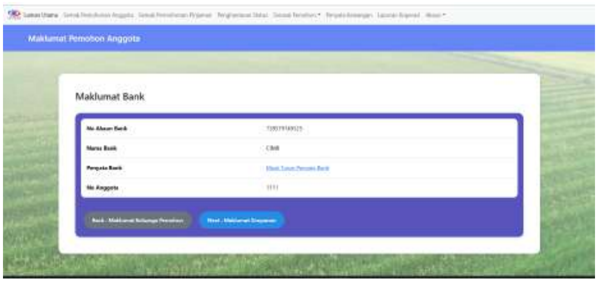
Review membership

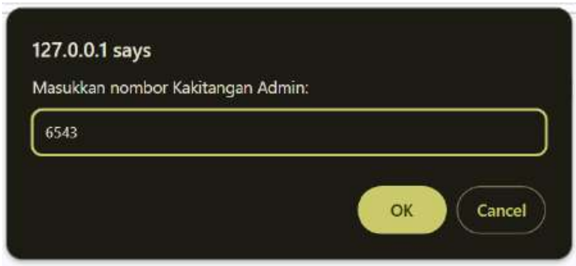








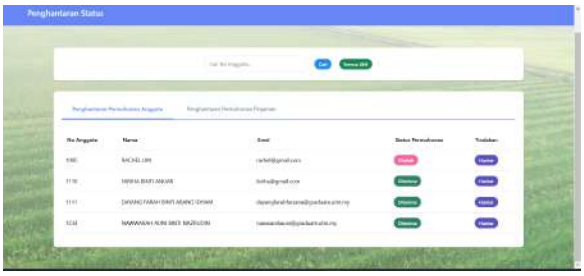




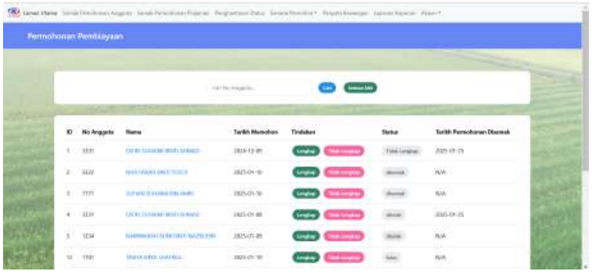

127.0.0.1 says
Nyatakan sebab keahlian tidak lengkap:
OK
Cancel


127.0.0.1 says
Status dikemaskini dengan jayanya.
OK

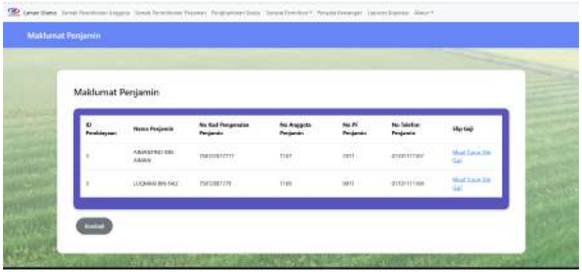

Koperasi Kakitangan KADA Kelantan
to me
Salam sejahtera, Safiya Masnoor.
Permohonan keahlian anda telah ditolak kerana:
PENYATA BANK TIDAK ADA NAMA BANK PEMILIK
Reply
Forward

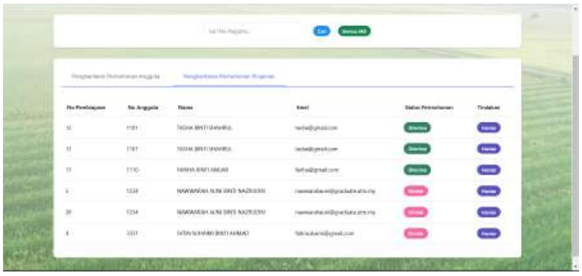

Penghantaran Status

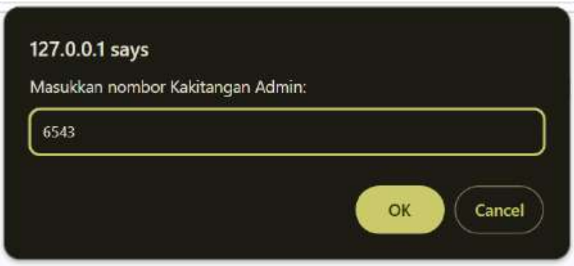
Review loan


Permohonan Pembiayaan


Maklumat Pemohon Pembiayaan
Maklumat Pembiayaan


Maklumat Penjamin
Maklumat Penjamin



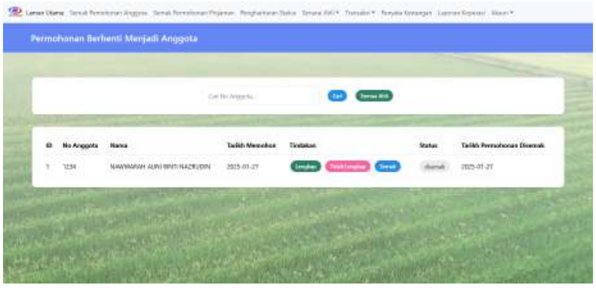

127.0.0.1 says
Masukkan nombor Kakitangan Admin:
6543
OK
Cancel

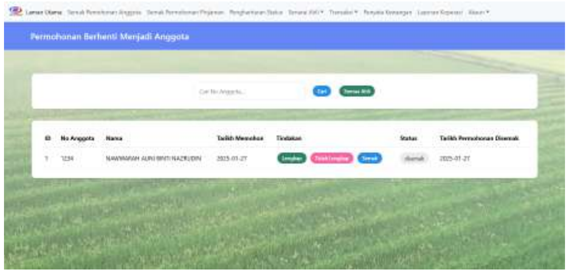

127.0.0.1 says
Nyatakan sebab keahlian tidak lengkap:
OK
Cancel

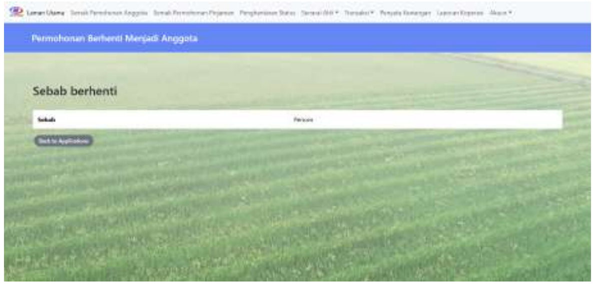

127.0.0.1 says
Status dikemaskini dengan jayanya.
OK

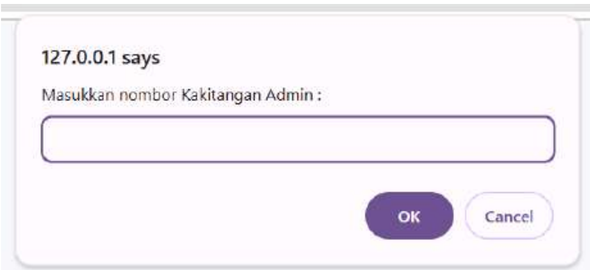

Koperasi Kakitangan KADA Kelantan
Salam sejahtera, Safiya Masnoor.
Permohonan keahlian anda telah ditolak kerana:
PENYATA BANK TIDAK ADA NAMA BANK PEMILIK
Reply
Forward

Review terminate application



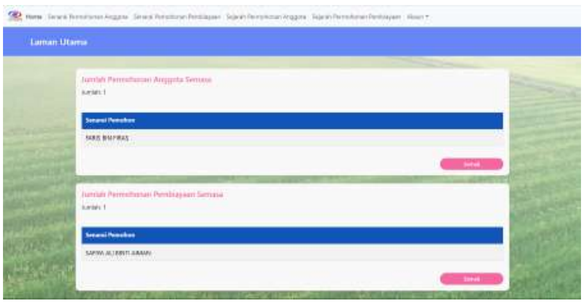


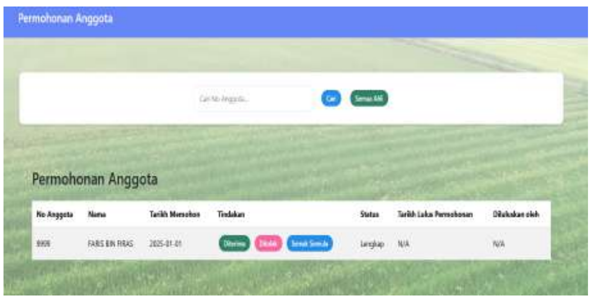


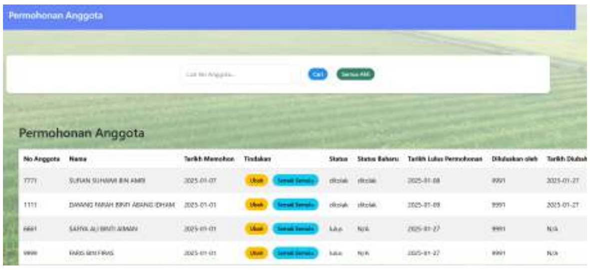


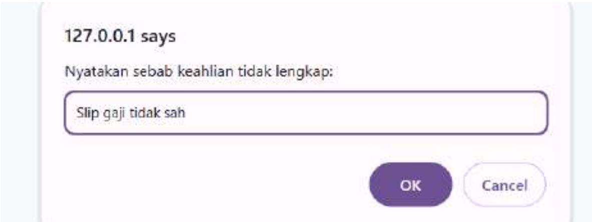
4.1.5. Approval Module

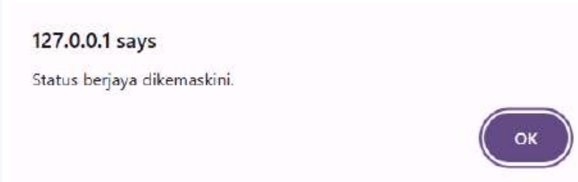
Approve Membership



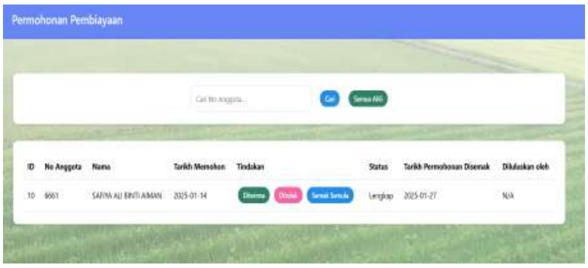


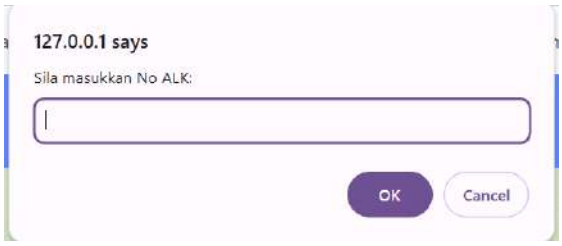


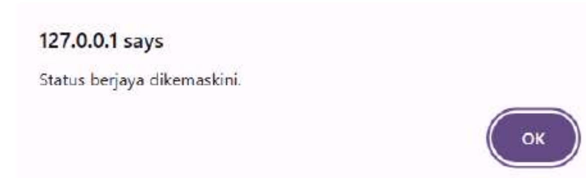


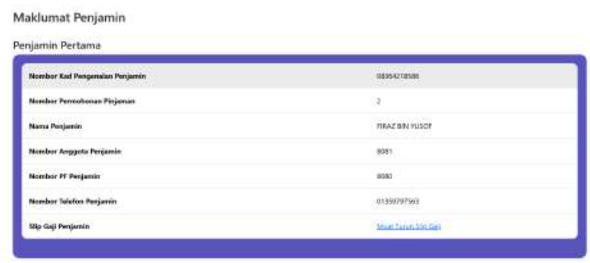


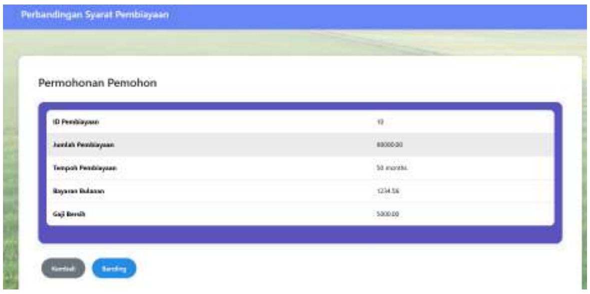
Approve Loan

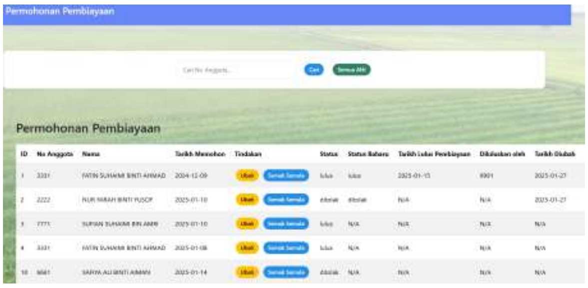




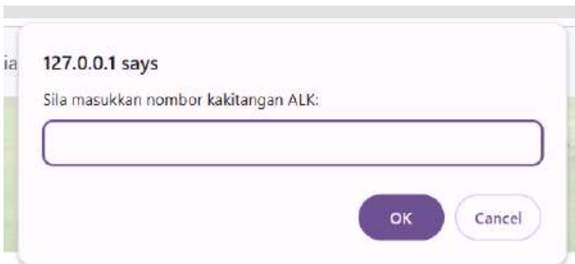




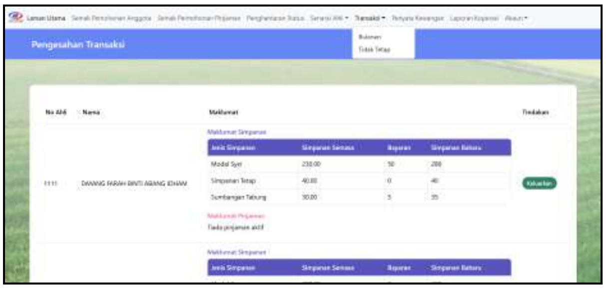


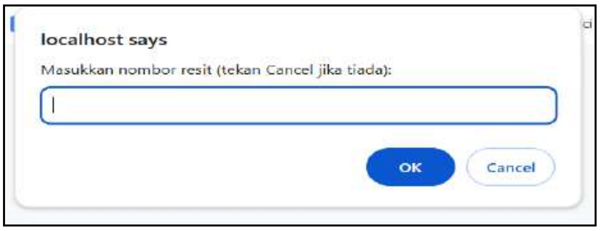


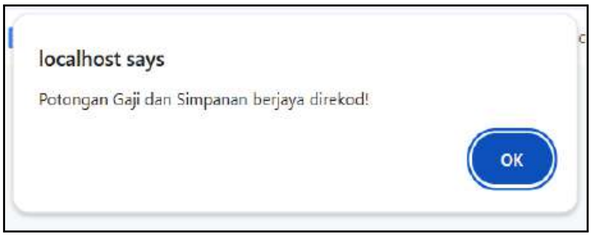
Approve terminate application

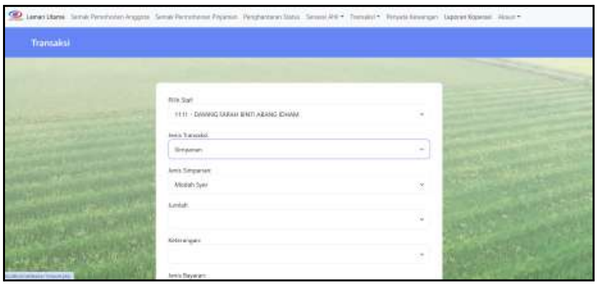


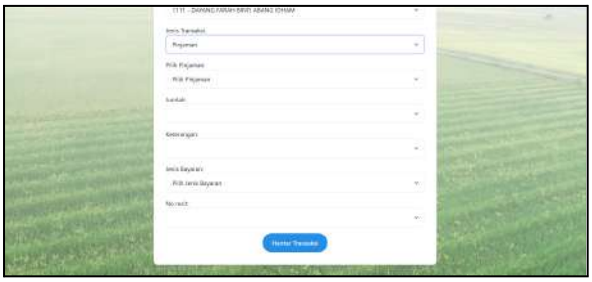
4.1.6. Financial Tracking Module



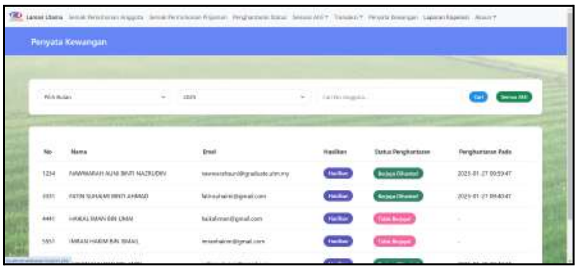


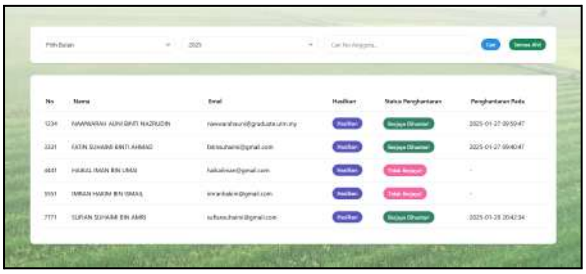






4.1.7. Statement Report Production Module


Penyata Kewangan

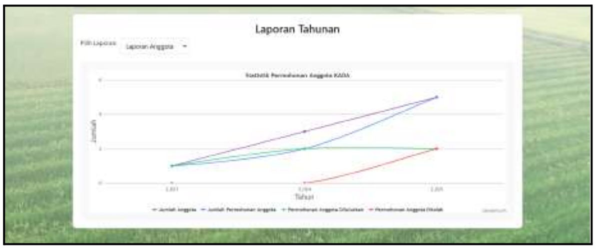


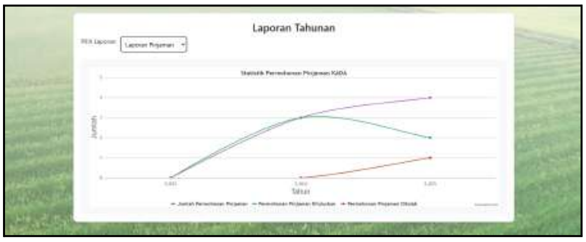

Penyata Kewangan

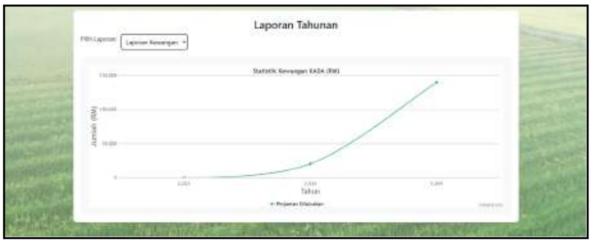

localhost says
Email Berjaya Dihantar!
OK
Modal Yuran:
Simpanan Tetap:
Tabun


Laporan Koperasi
RM 140,000.00
Laporan Bulanan
Statistik Keahlian
Statistik Pinjaman

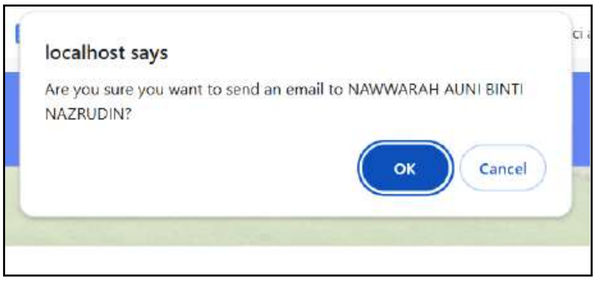

Laporan Bulanan
Statistik Keahlian
Statistik Pinjaman
Laporan Tahunan

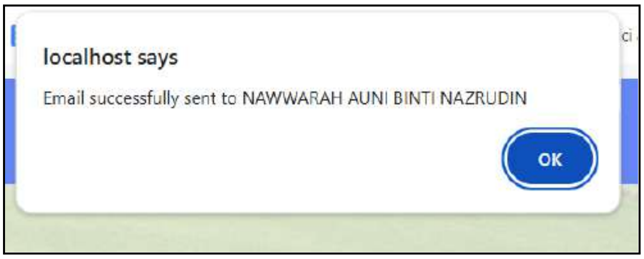

Laporan Tahunan

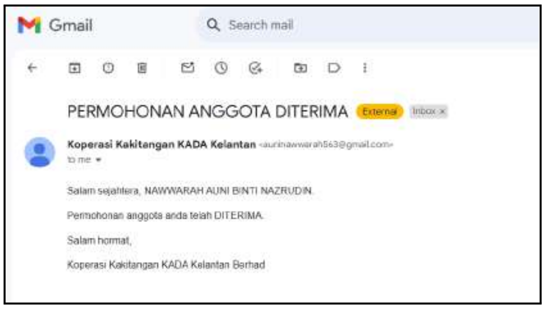

Laporan Tahunan

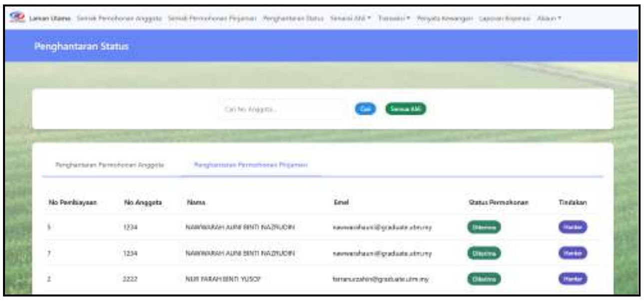

Laporan Tahunan

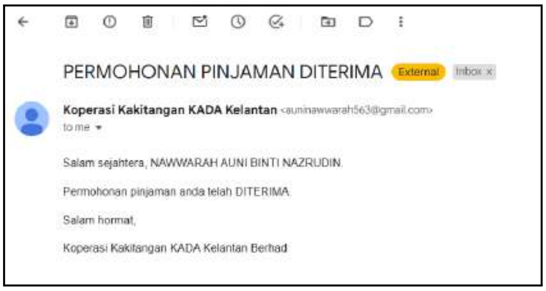

Penghantaran Status


localhost says
Are you sure you want to send an email to NAWWARAH AUNI BINTI NAZRUDIN?
OK
Cancel









4.2. SQL Statement (DDL & DML)

This section shows the Data Definition Language (DDL) and Data Manipulation Language (DML) used in the proposed system. DDL is used to define, create and modify the structure of database objects such as tables and columns. DML is used to manage and manipulate the data stored in the database. It includes operations like inserting, updating and selecting data.

4.2.1. Data Definition Language (DDL)

1. CREATE TABLE applicant

```sql
CREATE TABLE `applicant` (
  `staffNo` int(5) NOT NULL,
  `photo` longblob NOT NULL,
  `applicantName` varchar(255) NOT NULL,
  `applicantIC` varchar(12) NOT NULL,
  `applicantPF` bigint(10) NOT NULL,
  `applicantStreet` varchar(200) NOT NULL,
  `applicantPostcode` int(6) NOT NULL,
  `applicantCity` varchar(100) NOT NULL,
  `applicantState` varchar(100) NOT NULL,
  `applicantGender` varchar(10) NOT NULL,
  `applicantReligion` varchar(20) NOT NULL,
  `applicantRace` varchar(20) NOT NULL,
  `applicantPosition` varchar(50) NOT NULL,
  `officeStreet` varchar(255) NOT NULL,
  `officePostcode` int(6) NOT NULL,
  `officeCity` varchar(100) NOT NULL,
  `applicantPhoneNumber` varchar(15) NOT NULL,
  `applicantDOB` date NOT NULL,
  `applicantAge` int(5) NOT NULL,
  `maritalStatus` varchar(15) NOT NULL,
  `applicantGrade` varchar(10) NOT NULL,
  `officeFax` varchar(15) NOT NULL,
  `applicantPhoneHome` varchar(15) NOT NULL,
  `applicantSalary` decimal(10,2) NOT NULL,
  `salarySatement` longblob NOT NULL,
  `email` varchar(50) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
☐	1	staffNo	int(5)			No	None			Change	Drop	More
☐	2	photo	longblob			No	None			Change	Drop	More
☐	3	applicantName	varchar(255)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	4	applicantIC	varchar(12)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	5	applicantPF	bigint(10)			No	None			Change	Drop	More
☐	6	applicantStreet	varchar(200)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	7	applicantPostcode	int(6)			No	None			Change	Drop	More
☐	8	applicantCity	varchar(100)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	9	applicantState	varchar(100)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	10	applicantGender	varchar(10)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	11	applicantReligion	varchar(20)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	12	applicantRace	varchar(20)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	13	applicantPosition	varchar(50)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	14	officeStreet	varchar(255)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	15	officePostcode	int(6)			No	None			Change	Drop	More
☐	16	officeCity	varchar(100)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	17	applicantPhoneNumber	varchar(15)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	18	applicantDOB	date			No	None			Change	Drop	More
☐	19	applicantAge	int(5)			No	None			Change	Drop	More
☐	20	maritalStatus	varchar(15)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	21	applicantGrade	varchar(10)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	22	officeFax	varchar(15)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	23	applicantPhoneHome	varchar(15)	utf8mb4_general_ci		No	None			Change	Drop	More
☐	24	applicantSalary	decimal(10,2)			No	None			Change	Drop	More
☐	25	salarySatement	longblob			No	None			Change	Drop	More
☐	26	email	varchar(50)	utf8mb4_general_ci		No	None			Change	Drop	More

2. CREATE TABLE familyinfo

```
CREATE TABLE `familyinfo` (
  `familyIC` varchar(20) NOT NULL,
  `relationship` varchar(25) NOT NULL,
  `familyName` varchar(255) NOT NULL,
  `staffNo` int(5) NOT NULL,
  `count` int(15) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	**familyIC**	varchar(20)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	2	**relationship**	varchar(25)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	3	**familyName**	varchar(255)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	4	**staffNo**	int(5)			No	*None*			Change Drop More
☐	5	**count**	int(15)			No	*None*			Change Drop More

3. CREATE TABLE bankinfo

```
CREATE TABLE `bankinfo` (
  `accountBankNumber` bigint(25) NOT NULL,
  `bankAccountName` varchar(40) NOT NULL,
  `bankStatement` longblob NOT NULL,
  `staffNo` int(5) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	**accountBankNumber**	bigint(25)			No	*None*			Change Drop More
☐	2	**bankAccountName**	varchar(40)	utf8mb4_general_ci		No	*None*			Change Drop More
☐	3	**bankStatement**	longblob			No	*None*			Change Drop More
☐	4	**staffNo**	int(5)			No	*None*			Change Drop More

4. CREATE TABLE savingtype

```
CREATE TABLE `savingtype` (
  `feeMasuk` decimal(10,0) NOT NULL,
  `modahSyer` decimal(10,0) NOT NULL,
  `modalYuran` decimal(10,2) NOT NULL,
  `wangDepositAnggota` decimal(10,2) NOT NULL,
  `sumbanganTabung` decimal(10,2) NOT NULL,
  `simpananTetap` decimal(10,2) NOT NULL,
  `lainLain` decimal(10,2) NOT NULL,
  `staffNo` int(5) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	feeMasuk	decimal(10,0)			No	None			Change Drop More
☐	2	modahSyer	decimal(10,0)			No	None			Change Drop More
☐	3	modalYuran	decimal(10,2)			No	None			Change Drop More
☐	4	wangDepositAnggota	decimal(10,2)			No	None			Change Drop More
☐	5	sumbanganTabung	decimal(10,2)			No	None			Change Drop More
☐	6	simpananTetap	decimal(10,2)			No	None			Change Drop More
☐	7	lainLain	decimal(10,2)			No	None			Change Drop More
☐	8	staffNo	int(5)			No	None			Change Drop More

5. CREATE TABLE admin

```
CREATE TABLE `admin` (
  `staffNo` int(5) NOT NULL,
  `email` varchar(50) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	staffNo	int(5)			No	None			Change Drop More
☐	2	email	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More

6. CREATE TABLE membership

```sql
CREATE TABLE `membership` (
  `membershipStatus` int(3) NOT NULL,
  `membershipApplyDate` date NOT NULL,
  `membershipApproveDate` date DEFAULT NULL,
  `membershipReviewDate` date DEFAULT NULL,
  `staffNo` int(5) NOT NULL,
  `alkStaffNo` int(5) DEFAULT NULL,
  `adminStaffNo` int(5) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
☐	1	**membershipStatus**	int(3)			No	*None*			Change	Drop	More
☐	2	**membershipApplyDate**	date			No	*None*			Change	Drop	More
☐	3	**membershipApproveDate**	date			Yes	*NULL*			Change	Drop	More
☐	4	**membershipReviewDate**	date			Yes	*NULL*			Change	Drop	More
☐	5	**staffNo**	int(5)			No	*None*			Change	Drop	More
☐	6	**alkStaffNo**	int(5)			Yes	*NULL*			Change	Drop	More
☐	7	**adminStaffNo**	int(5)			Yes	*NULL*			Change	Drop	More

7. CREATE TABLE status

```
CREATE TABLE `status` (
  `status` int(3) NOT NULL PRIMARY KEY,
  `statusDesc` varchar(25) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

```
INSERT INTO `status` (`status`, `statusDesc`) VALUES
(1, 'disemak'),
(2, 'lulus'),
(3, 'ditolak'),
(4, 'Tidak Lengkap'),
(5, 'Lengkap');
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**status**	int(3)			No	*None*			Change Drop More
2	**statusDesc**	varchar(25)	utf8mb4_general_ci		No	*None*			Change Drop More

8. CREATE TABLE loan

```
CREATE TABLE `loan` (
  `loanID` int(5) NOT NULL PRIMARY KEY,
  `loanAmount` decimal(10,2) NOT NULL,
  `loanDuration` int(2) NOT NULL,
  `loanStatus` int(3) NOT NULL,
  `monthlyPayment` decimal(10,2) NOT NULL,
  `bankName` varchar(30) NOT NULL,
  `accountBankNumber` bigint(25) NOT NULL,
  `applicantNetSalary` decimal(10,2) NOT NULL,
  `applicantBasicSalary` decimal(10,2) NOT NULL,
  `pengesahanMajikan` longblob NOT NULL,
  `loanApproveDate` date DEFAULT NULL,
  `staffNo` int(5) NOT NULL,
  `loanType` int(7) NOT NULL,
  `otherLoan` varchar(25) DEFAULT NULL,
  `loanReviewDate` date DEFAULT NULL,
  `loanApplyDate` date NOT NULL,
  `adminStaffNo` int(5) DEFAULT NULL,
  `alkStaffNo` int(5) DEFAULT NULL,
 FOREIGN KEY (loanStatus)
   REFERENCES status (status),
 FOREIGN KEY (staffNo)
   REFERENCES applicant (staffNo),
 FOREIGN KEY (loanType)
   REFERENCES loantype (loanType),
 FOREIGN KEY (adminstaffNo)
   REFERENCES admin (staffNo),
 FOREIGN KEY (alkstaffNo)
   REFERENCES alk (staffNo)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
1	loanID	int(5)			No	None		AUTO_INCREMENT	Change	Drop	More
2	loanAmount	decimal(10,2)			No	None			Change	Drop	More
3	loanDuration	int(2)			No	None			Change	Drop	More
4	loanStatus	int(3)			No	None			Change	Drop	More
5	monthlyPayment	decimal(10,2)			No	None			Change	Drop	More
6	bankName	varchar(30)	utf8mb4_general_ci		No	None			Change	Drop	More
7	accountBankNumber	bigint(25)			No	None			Change	Drop	More
8	applicantNetSalary	decimal(10,2)			No	None			Change	Drop	More
9	applicantBasicSalary	decimal(10,2)			No	None			Change	Drop	More
10	pengesahanMajikan	longblob			No	None			Change	Drop	More
11	loanApproveDate	date			Yes	NULL			Change	Drop	More
12	staffNo	int(5)			No	None			Change	Drop	More
13	loanType	int(7)			No	None			Change	Drop	More
14	otherLoan	varchar(25)	utf8mb4_general_ci		Yes	NULL			Change	Drop	More
15	loanReviewDate	date			Yes	NULL			Change	Drop	More
16	loanApplyDate	date			No	None			Change	Drop	More
17	adminStaffNo	int(5)			Yes	NULL			Change	Drop	More
18	alkStaffNo	int(5)			Yes	NULL			Change	Drop	More

9. CREATE TABLE loantype

```
CREATE TABLE `loantype` (
  `loanType` int(7) NOT NULL PRIMARY KEY,
  `loanName` varchar(30) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	**loanType**	int(7)			No	*None*			Change Drop More
☐	2	**loanName**	varchar(30)	utf8mb4_general_ci		No	*None*			Change Drop More

10. CREATE TABLE ALK

```
CREATE TABLE `alk` (
  `staffNo` int(5) NOT NULL,
  `email` varchar(50) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**staffNo**	int(5)			No	*None*			Change
2	**email**	varchar(50)	utf8mb4_general_ci		No	*None*			Change

11. ALTER TABLE ALK

```
ALTER TABLE `alk`
  ADD PRIMARY KEY (`staffNo`),
  ADD UNIQUE KEY `email` (`email`);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**staffNo**	int(5)			No	*None*			Change
2	**email**	varchar(50)	utf8mb4_general_ci		No	*None*			Change

12. CREATE TABLE Guarantorinfo

```sql
CREATE TABLE `guarantorinfo` (
  `guarantorIC` varchar(12) NOT NULL,
  `guarantorName` varchar(100) NOT NULL,
  `guarantorID` int(5) NOT NULL,
  `guarantorPF` int(5) NOT NULL,
  `guarantorPhoneNumber` varchar(11) NOT NULL,
  `photo` longblob NOT NULL,
  `loanID` int(5) NOT NULL,
  PRIMARY KEY (guarantorIC, loanID)
  FOREIGN KEY (loanID)
    REFERENCES loan (loanID)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	guarantorIC	varchar(12)	utf8mb4_general_ci		No	*None*			Change Drop More
2	guarantorName	varchar(100)	utf8mb4_general_ci		No	*None*			Change Drop More
3	guarantorID	int(5)			No	*None*			Change Drop More
4	guarantorPF	int(5)			No	*None*			Change Drop More
5	guarantorPhoneNumber	varchar(11)	utf8mb4_general_ci		No	*None*			Change Drop More
6	photo	longblob			No	*None*			Change Drop More
7	loanID	int(5)			No	*None*			Change Drop More

13. CREATE TABLE Application Table

```sql
CREATE TABLE Application (
    staffNo INT(5) NOT NULL,
    loanID INT(20),
    membershipStatus INT(5),
    loanStatus INT(5),
    membershipApplyDate DATE NOT NULL,
    loanApplyDate DATE,
    PRIMARY KEY (staffNo, loanID),
    FOREIGN KEY (staffNo) REFERENCES Applicant(staffNo),
    FOREIGN KEY (loanID) REFERENCES Loan(loanID)
);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	staffNo	int(5)			No	*None*			Change Drop More
2	loanID	int(20)			No	*None*			Change Drop More
3	membershipStatus	int(5)			Yes	*NULL*			Change Drop More
4	loanStatus	int(5)			Yes	*NULL*			Change Drop More
5	membershipApplyDate	date			No	*None*			Change Drop More
6	loanApplyDate	date			Yes	*NULL*			Change Drop More

14. CREATE TABLE historymodiloan

```
CREATE TABLE `historymodiloan` (
  `historyLoanID` int(5) NOT NULL COMMENT 'Unique identifier for each past application',
  `newLoanStatus` int(3) DEFAULT NULL COMMENT 'new status after modificatiion',
  `loanChangeDate` date DEFAULT NULL COMMENT 'date status changed',
  `loanModifiedBy` int(5) DEFAULT NULL COMMENT 'modifier''s staff number',
  `loanID` int(5) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
historyLoanID	int(5)			No	None	Unique identifier for each past application	AUTO_INCREMENT	Change Drop More
newLoanStatus	int(3)			Yes	NULL	new status after modification		Change Drop More
loanChangeDate	date			Yes	NULL	date status changed		Change Drop More
loanModifiedBy	int(5)			Yes	NULL	modifier's staff number		Change Drop More
loanID	int(5)			No	None			Change Drop More

15. CREATE TABLE historymodimem

```
CREATE TABLE `historymodimem` (
  `historyMemID` int(5) NOT NULL COMMENT 'Unique identifier for each past application',
  `newMemStatus` int(3) DEFAULT NULL COMMENT 'New status after modification',
  `memChangeDate` date DEFAULT NULL COMMENT 'Date status modifed',
  `memModifiedBy` int(5) DEFAULT NULL COMMENT 'Modifier''s staff number',
  `membershipID` int(5) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
historyMemID	int(5)			No	None	Unique identifier for each past application	AUTO_INCREMENT	Change Drop More
newMemStatus	int(3)			Yes	NULL	New status after modification		Change Drop More
memChangeDate	date			Yes	NULL	Date status modififed		Change Drop More
memModifiedBy	int(5)			Yes	NULL	Modifier's staff number		Change Drop More
membershipID	int(5)			No	None			Change Drop More

16. CREATE TABLE alkapplicationfeedback

```
CREATE TABLE `alkapplicationfeedback` (
  `feedbackID` int(20) NOT NULL,
  `email` varchar(30) NOT NULL,
  `reason` longtext NOT NULL,
  `dateGiven` date NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	feedbackID	int(20)			No	None		AUTO_INCREMENT	Change Drop More
2	email	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
3	reason	longtext	utf8mb4_general_ci		No	None			Change Drop More
4	dateGiven	date			No	None			Change Drop More

17. CREATE TABLE, ALTER TABLE…ADD Managing table

```sql
CREATE TABLE `managing` (
  `status` varchar(25) DEFAULT NULL,
  `notificationLatestDate` varchar(25) DEFAULT NULL,
  `email` varchar(50) NOT NULL,
  `staffNo` int(5) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

```sql
ALTER TABLE `managing`
  ADD UNIQUE KEY `email` (`email`),
  ADD UNIQUE KEY `staffNo` (`staffNo`);
```

```sql
ALTER TABLE `managing`
  ADD CONSTRAINT `managing_ibfk_2` FOREIGN KEY (`staffNo`) REFERENCES `admin` (`staffNo`),
  ADD CONSTRAINT `managing_ibfk_3` FOREIGN KEY (`email`) REFERENCES `users` (`email`);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	status	varchar(25)	utf8mb4_general_ci		Yes	NULL			Change Drop More
2	notificationLatestDate	varchar(25)	utf8mb4_general_ci		Yes	NULL			Change Drop More
3	email	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
4	staffNo	int(5)			Yes	NULL			Change Drop More

18. CREATE TABLE, ALTER TABLE…ADD Saving table

```sql
CREATE TABLE `loanrepayment` (
  `repaymentID` int(5) NOT NULL,
  `repaymentAmount` decimal(10,2) NOT NULL,
  `repaymentDate` date NOT NULL,
  `repaymentDesc` varchar(200) DEFAULT NULL,
  `repaymentType` varchar(30) NOT NULL,
  `repaymentReceipt` varchar(15) DEFAULT NULL,
  `repaymentConfirm` varchar(15) DEFAULT NULL,
  `loanID` int(5) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

```sql
ALTER TABLE `loanrepayment`
  ADD PRIMARY KEY (`repaymentID`),
  ADD KEY `loanID` (`loanID`);
```

```sql
ALTER TABLE `saving`
  ADD CONSTRAINT `saving_ibfk_1` FOREIGN KEY (`staffNo`) REFERENCES `applicant` (`staffNo`);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	savingID	int(5)			No	*None*		AUTO_INCREMENT	Change Drop More
2	savingAmount	decimal(10,2)			No	*None*			Change Drop More
3	savingDate	date			No	*None*			Change Drop More
4	savingDesc	varchar(200)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
5	savingType	varchar(30)	utf8mb4_general_ci		No	*None*			Change Drop More
6	savingReceipt	varchar(15)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
7	savingConfirm	varchar(15)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
8	staffNo	int(5)			No	*None*			Change Drop More

19. CREATE TABLE, ALTER TABLE…ADD loanRepayment table

```
CREATE TABLE `saving` (
  `savingID` int(5) NOT NULL,
  `savingAmount` decimal(10,2) NOT NULL,
  `savingDate` date NOT NULL,
  `savingDesc` varchar(200) DEFAULT NULL,
  `savingType` varchar(30) NOT NULL,
  `savingReceipt` varchar(15) DEFAULT NULL,
  `savingConfirm` varchar(15) DEFAULT NULL,
  `staffNo` int(5) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

```
ALTER TABLE `saving`
  ADD PRIMARY KEY (`savingID`),
  ADD KEY `staffNo` (`staffNo`);
```

```
ALTER TABLE `loanrepayment`
  ADD CONSTRAINT `loanrepayment_ibfk_1` FOREIGN KEY (`loanID`) REFERENCES `loan` (`loanID`);
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	repaymentID	int(5)			No	*None*		AUTO_INCREMENT	Change Drop More
2	repaymentAmount	decimal(10,2)			No	*None*			Change Drop More
3	repaymentDate	date			No	*None*			Change Drop More
4	repaymentDesc	varchar(200)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
5	repaymentType	varchar(30)	utf8mb4_general_ci		No	*None*			Change Drop More
6	repaymentReceipt	varchar(15)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
7	repaymentConfirm	varchar(15)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
8	loanID	int(5)			No	*None*			Change Drop More

4.2.2. Data Manipulation Language (DML)

4.2.2.1. Membership Application Module DML

1. INSERT INTO applicant

The INSERT INTO function is to insert the applicant information into the database.

```
$sql = "INSERT INTO applicant (staffNo, applicantName, applicantIC, applicantPF, applicantStreet, applicantPostcode, applicantCity
    , applicantState, applicantGender, applicantReligion, applicantRace, applicantPosition, officeStreet, officePostcode,
    officeCity, applicantPhoneNumber, applicantDOB, applicantAge, maritalStatus, applicantGrade, officeFax, applicantPhoneHome,
    applicantSalary, email, photo, salaryStatement)
        VALUES ('$sNo', '$aName', '$aIC', '$aPF', '$aStreet','$aPostcode','$aCity','$aState','$aGender','$aReligion','$aRace','$
            aPosition','$oStreet','$oPostcode','$oCity','$aPhoneNo','$aDOB','$aAge','$maritalStat','$aGrade','$oFax','$aPhoneHome'
            ,'$aSalary', '$femail', '$photoPath', '$salaryStatementPath')";
```

staffNo	photo	applicantName 1	applicantIC	applicantPF	applicantStreet	applicantPostcode	applicantCity	applicantState	applicantGender
4433	[BLOB - 98.8 KiB]	aamin	040305120093	3344	NO15 JALAN KANTANSARI 3	48010	RAWANG	SELANGOR	LELAKI

applicantReligion	applicantRace	applicantPosition	officeStreet	officePostcode	officeCity	applicantPhoneNumber	applicantDOB
ISLAM	MELAYU	KETUA	NO 11A JALAN KEMBOJASARI	48010	RAWANG	0175556666	2004-04-03

applicantAge	maritalStatus	applicantGrade	officeFax	applicantPhoneHome	applicantSalary	salarySatement	email
21	BUJANG	G41	0360214456		1234.00	[BLOB - 98.8 KiB]	kang@gmail.com

2. INSERT INTO familyinfo

The INSERT INTO function is to insert the family information into the database.

```
$sql = "INSERT INTO familyinfo (staffNo, count, relationship, familyName, familyIC)
        VALUES ('$sNo', '$fCount', '$relationship', '$name', '$ic_number')";
```

familyIC	relationship 1	familyName	staffNo	count
710204040090	IBU	ALIA	4433	1

3. INSERT INTO bankinfo

The INSERT INTO function is to insert the bank information into the database.

```
$sql = "INSERT INTO bankinfo (accountBankNumber, bankAccountName, staffNo, bankStatement)
        VALUES ('$aBankNo', '$bName', '$sNo', '$newFileName')";
```

accountBankNumber	bankAccountName ▲ 1	bankStatement	staffNo
11223344678	AFFIN BANK	[BLOB - 98.8 KiB]	4433

4. INSERT INTO savingtype

The INSERT INTO function is to insert the savingtype into the database.

```
$sql = "INSERT INTO savingtype (staffNo, feeMasuk, modahSyer, modalYuran, wangDepositAnggota, sumbanganTabung, simpananTetap,
    lainLain)
        VALUES ('$sNo', '$fMasuk', '$mSyer', '$mYuran', '$wangDepo', '$sumTabung', '$simTetap', '$dll')";
```

feeMasuk ▲ 1	modahSyer	modalYuran	wangDepositAnggota	sumbanganTabung	simpananTetap	lainLain	staffNo
1	20	30.00	21.00	11.00	21.00	90.00	4433

4.2.2.2. Loan Application Module DML

1. SELECT membership.membershipStatus, savingtype.modahSyer

The SELECT function is to select a member that is already being approved and total modahSyer more than rm300.

```
$sqlMembership = "SELECT membership.membershipStatus, savingtype.modahSyer
                  FROM membership
                  JOIN applicant ON membership.staffNo = applicant.staffNo
                  JOIN savingtype ON membership.staffNo = savingType.staffNo
                  WHERE applicant.email = '$userEmail'";
```

membershipStatus	modahSyer
2	300

2. SELECT applicant.*, bankinfo.bankAccountName, bankinfo.accountBankNumber

This SELECT function is to select some of applicant information and bankAccountName as well as accountBankNumber from bankinfo table.

```
$sql = "SELECT applicant.*, bankinfo.bankAccountName, bankinfo.accountBankNumber
        FROM applicant
        LEFT JOIN users ON users.email = applicant.email
        LEFT JOIN bankinfo ON bankinfo.staffNo = applicant.staffNo
        LEFT JOIN membership ON membership.staffNo = applicant.staffNo
        WHERE users.email = '$userEmail'";
```

applicantName	applicantIC	applicantDOB	applicantAge	applicantStreet	applicantPostcode	applicantGender	applicantReligion	applicantRace	applicantPF	applicantPosition
NAWWARAH AUNI BINTI NAZRUDIN	40518160909	2004-05-18	21	JALAN KEMUBU 8/9	40320	P	ISLAM	MELAYU	5468	D

officeStreet	officePostcode	applicantPhoneNumber	bankAccountName	accountBankNumber
JALAN KEMUBU 8/9	40320	195143276	BANK RAKYAT	6472195734

3. INSERT INTO loan

The INSERT INTO function is to insert the loan application information filled in by the applicant.

```
$sqlInsertLoan = "INSERT INTO loan
        (loanAmount, loanDuration, loanStatus, monthlyPayment, bankName, accountBankNumber, loanType, otherLoan,
            staffNo, adminStaffNo, loanApplyDate, loanReviewDate, loanApproveDate, applicantNetSalary,
            applicantBasicSalary, pengesahanMajikan)
        VALUES ('$amount', '$duration', '1', '$mpayment', '$bank', '$accNum', '$type', '$other', '$staffNo', $
            adminStaffNo, CURRENT_TIMESTAMP(), $loanReviewDate, $loanApproveDate, '$netSalary', '$basicSalary', '
            $fileContent3')";
```

loanID	loanAmount	loanDuration	loanStatus	monthlyPayment	bankName	accountBankNumber	applicantNetSalary
20	10000.00	24	1	434.17	RHB Bank	76408751157	7511.00

applicantBasicSalary	pengesahanMajikan	loanApproveDate	staffNo	loanType	otherLoan	loanReviewDate	loanApplyDate	adminStaffNo	alkStaffNo
7890.00	[BLOB - 13 B]	NULL	1234	1	NULL	NULL	2025-01-21	NULL	NULL

4. INSERT INTO guarantorinfo

The INSERT INTO function is to insert the two of guarantor information when applying for a loan.

```
$sqlInsertGuarantor1 = "INSERT INTO guarantorInfo
                        (guarantorIC, loanID, guarantorName, guarantorID, guarantorPF, guarantorPhoneNumber, photo)
                        VALUES ('$gIC1', '$loanID', '$gName1', '$gID1', '$gPF1', '$gPhoneNumber1', '$fileContent1')";
```

```
$sqlInsertGuarantor2 = "INSERT INTO guarantorInfo
                        (guarantorIC, loanID, guarantorName, guarantorID, guarantorPF, guarantorPhoneNumber, photo)
                        VALUES ('$gIC2', '$loanID', '$gName2', '$gID2', '$gPF2', '$gPhoneNumber2', '$fileContent2')
                            ";
```

guarantorIC	guarantorName	guarantorID	guarantorPF	guarantorPhoneNumber	photo	loanID
710125101236	Qawie binti Awie	1236	6321	01666632222	[BLOB - 13 B]	20
751206012345	Muhammad bin Ahmad	3457	7543	01777754444	[BLOB - 13 B]	20

5. SELECT loanID, loanStatus, loanApplyDate, loanApproveDate, loan.loanType, loantype.loanType, loantype.loanName

This SELECT function is to select some of the loan information to be used to update the status of the loan application.

```
$query = "SELECT loanID, loanStatus, loanApplyDate, loanApproveDate, loan.loanType, loantype.loanType, loantype.loanName
        FROM loan
        LEFT JOIN applicant ON loan.staffNo = applicant.staffNo
        LEFT JOIN loantype ON loantype.loanType = loan.loanType
        WHERE applicant.email = '$userEmail'
        ORDER BY loanApplyDate DESC";
```

4.2.2.3. Terminate Module DML

1. SELECT membership.membershipStatus, savingtype.modahSyer

This SELECT function is to select membershipStatus and modahSyer from their respective table to make sure members are eligible to filled in the termination member form

```
$sqlMembership = "SELECT membership.membershipStatus, savingtype.modahSyer
                  FROM membership
                  JOIN applicant ON membership.staffNo = applicant.staffNo
                  JOIN savingtype ON membership.staffNo = savingType.staffNo
                  WHERE applicant.email = '$userEmail'";
```

2. SELECT applicant.*, bankinfo.bankAccountName, bankinfo.accountBankNumber

This SELECT function is to select applicant information to display in the form to ease users to not fill in their information again.

```
$sql = "SELECT applicant.*, bankinfo.bankAccountName, bankinfo.accountBankNumber
        FROM applicant
        LEFT JOIN users ON users.email = applicant.email
        LEFT JOIN bankinfo ON bankinfo.staffNo = applicant.staffNo
        LEFT JOIN membership ON membership.staffNo = applicant.staffNo
        WHERE users.email = '$userEmail'";
```

3. SELECT staffNo

This SELECT function is to select the staff number of applicants.

```
$sql = "SELECT staffNo
            FROM applicant
            JOIN users ON users.email = applicant.email
            WHERE users.email = '$userEmail'";
```

4. INSERT INTO membershipEnd

This INSERT INTO function is to insert the terminate member form information into membershipend table.

```
$insertQuery = "INSERT INTO membershipend (staffNo, applyDate, endDate, reason, reviewDate, approveDate, adminStaffNo,
    alkStaffNo, status)
                VALUES ('$staffNo', CURRENT_TIMESTAMP(), $endDate, $reason, $reviewDate, $approveDate, $adminStaffNo, $
                    alkStaffNo, '1')";
```

4.2.2.4. Review Module DML

Review Membership

1. SELECT staffNo, applicantName, membershipApplyDate, membershipStatus, membershipReviewDate, statusDesc

This function is to select staffNo, applicantName, membershipApplyDate, membershipStatus, membershipReviewDate from the applicant table and statusDesc from status table.

```
$sql = "SELECT
            applicant.staffNo,
            applicant.applicantName,
            membership.membershipApplyDate,
            membership.membershipStatus,
            membership.membershipReviewDate,
            status.statusDesc
        FROM membership
        LEFT JOIN applicant ON membership.staffNo = applicant.staffNo
        LEFT JOIN status ON membership.membershipStatus = status.status";
```

staffNo	applicantName	membershipApplyDate	membershipStatus	membershipReviewDate	statusDesc
1005	RACHEL LIM	2025-01-19	1	NULL	disemak
1101	TASHA BINTI SHAHRUL	2025-01-19	1	NULL	disemak
1110	FARIHA BINTI ANUAR	2025-01-19	2	2025-01-19	lulus
1111	DAYANG FARAH BINTI ABANG IDHAM	2025-01-09	2	NULL	lulus
1234	NAWWARAH AUNI BINTI NAZRUDIN	2025-02-09	2	NULL	lulus
2222	NUR FARAH BINTI YUSOF	2025-03-09	1	NULL	disemak
3331	FATIN SUHAIMI BINTI AHMAD	2024-04-09	1	NULL	disemak
4441	HAIKAL IMAN BIN UMAI	2023-07-09	1	NULL	disemak
5551	IMRAN HAKIM BIN ISMAIL	2024-08-09	1	NULL	disemak
6661	SAFIYA ALI BINTI AIMAN	2025-01-09	1	NULL	disemak
7771	SUFIAN SUHAIMI BIN AMRI	2025-01-09	1	NULL	disemak

2. SELECT staffNo, applicantName, applicantIC, applicantPF, applicantStreet, applicantPostcode, applicantCity, applicantState, applicantGender, applicantReligion, applicantRace, applicantPosition, officeStreet, officePostcode, officeCity, applicantPhoneNumber, applicantDOB, applicantAge, maritalStatus, applicantGrade, officeFax, applicantPhoneHome, applicantSalaray, photo, salaryStatement

This function is to select applicant information from the applicant table to be displayed in membership form.

```
$sql = "SELECT
        staffNo, applicantName, applicantIC, applicantPF, applicantStreet,
        applicantPostcode, applicantCity, applicantState, applicantGender,
        applicantReligion, applicantRace, applicantPosition, officeStreet,
        officePostcode, officeCity, applicantPhoneNumber, applicantDOB,
        applicantAge, maritalStatus, applicantGrade, officeFax,
        applicantPhoneHome, applicantSalary, photo, salaryStatement
    FROM applicant
    WHERE staffNo = ?";
```

staffNo	applicantName	applicantIC	applicantPF	applicantStreet	applicantPostcode	applicantCity	applicantState	applicantGender
1110	FARIHA BINTI ANUAR	0828093042	1011	NO.89, JALAN NUMI 1/2	47100	KUALA LIPIS	Selangor	PEREMPUAN

applicantReligion	applicantRace	applicantPosition	officeStreet	officePostcode
Islam	Melayu	STAFF	LOT 10	47100

officeCity	applicantPhoneNumber	applicantDOB	applicantAge	maritalStatus	applicantGrade	officeFax	applicantPhoneHome	applicantSalary
KUALA LIPIS	0156789876	2025-01-04	38	Berkahwin	G14	0301123443	0300000009	5000.00

3. SELECT count, relationship, familyName, familyIC, staffNo

This function is to select family information from the familyinfo table to be displayed in membership form.

```
$sql = "SELECT
        count, relationship, familyName, familyIC, staffNo
    FROM familyinfo
    WHERE staffNo = ?";
```

count	relationship	familyName	familyIC	staffNo
2	ADIK-BERADIK	SHAHRUL BIN FARIS	209874805070	1110
2	ANAK-BERANAK	FIRAS BIN SARIF	87940720983	1110

4. SELECT accountBankNumber, bankAccountName, staffNo, bankStatement

This function is to select bank information from the bankinfo table to be displayed in membership form.

```
$sql = "SELECT
        accountBankNumber, bankAccountName, staffNo, bankStatement
    FROM bankinfo
    WHERE staffNo = ?";
```

accountBankNumber	bankAccountName	staffNo
2222222222222	CIMB Group Holdings	1110

5. SELECT staffNo, feeMasuk, modahSyer, modalYuran, wangDepositAnggota, sumbanganTabung, simpananTetap, lainLain

This function is to select some of saving type information from savingtype table to be displayed in membership form

```
$sql = "SELECT
        staffNo, feeMasuk, modahSyer, modalYuran, wangDepositAnggota, sumbanganTabung, simpananTetap, lainLain
    FROM savingtype
    WHERE staffNo = ?";
```

staffNo	feeMasuk	modahSyer	modalYuran	wangDepositAnggota	sumbanganTabung	simpananTetap	lainLain
1110	50	100	30.00	55.00	60.00	10.00	0.00

6. UPDATE membership status

Update the membership table with the provided adminStaffNo.

```
$sql = "UPDATE membership
        SET membershipStatus = $status,
            membershipReviewDate = '$reviewDate',
            adminStaffNo = " . ($adminStaffNo ? "'$adminStaffNo'" : "NULL") .
        WHERE staffNo = '$staffNo'";
```

membershipStatus	membershipApplyDate	membershipReviewDate
2	2025-01-19	2025-01-19

7. SELECT email, firstName,LastName.

This function is to select email, first name and last name from the applicant table.

```
SELECT u.email, u.firstName, u.lastName
       FROM users u
       JOIN applicant a ON a.email = u.email
       WHERE a.staffNo = '6661';
```

email	firstName	lastName
safiyamasnoor@gmail.com	Safiya	Masnoor

8. INSERT INTO ApplicationFeedback table for membership application

```
INSERT INTO ApplicationFeedback (email, reason, dateGiven)
VALUES ('safiyamasnoor@gmail.com', 'PENYATA BANK TIDAK ADA NAMA PEMILIK BANK', CURDATE());
```

←T→	feedbackID	email	reason	dateGiven
Edit Copy Delete	25	safiyamasnoor@gmail.com	PENYATA BANK TIDAK ADA NAMA PEMILIK BANK	2025-01-26

Review Loan

1. SELECT staffNo, applicantName, loanID, loanApplyDate, loanStatus, loanReviewDate, status Desc

This function is to fetch the required data and join with the status table.

```
$sql = "SELECT
            applicant.staffNo,
            applicant.applicantName,
            loan.loanID,
            loan.loanApplyDate,
            loan.loanStatus,
            loan.loanReviewDate,
            status.statusDesc
        FROM loan
        LEFT JOIN applicant ON loan.staffNo = applicant.staffNo
        LEFT JOIN status ON loan.loanStatus = status.status"; //
```

staffNo	applicantName	loanID	loanApplyDate	loanStatus	loanReviewDate	statusDesc
3331	FATIN SUHAIMI BINTI AHMAD	1	2024-12-09	1	NULL	disemak
2222	NUR FARAH BINTI YUSOF	2	2025-01-10	1	NULL	disemak
7771	SUFIAN SUHAIMI BIN AMRI	3	2025-01-10	1	NULL	disemak
3331	FATIN SUHAIMI BINTI AHMAD	4	2025-01-08	1	NULL	disemak
1234	NAWWARAH AUNI BINTI NAZRUDIN	5	2025-01-09	2	NULL	lulus
6661	SAFIYA ALI BINTI AIMAN	6	2025-01-16	1	NULL	disemak
1101	TASHA BINTI SHAHRUL	12	2025-01-19	2	NULL	lulus
1101	TASHA BINTI SHAHRUL	13	2025-01-19	2	2025-01-19	lulus
1101	TASHA BINTI SHAHRUL	14	2025-01-20	1	NULL	disemak
1101	TASHA BINTI SHAHRUL	15	2025-01-20	1	NULL	disemak
1110	FARIHA BINTI ANUAR	17	2025-01-20	2	2025-01-20	lulus
1110	FARIHA BINTI ANUAR	18	2025-01-20	4	2025-01-21	Tidak Lengkap
1234	NAWWARAH AUNI BINTI NAZRUDIN	20	2025-01-21	1	NULL	disemak

2. SELECT loanID, loanAmount, loanDuration, monthlyPayment, bankName, accountBankNumber, loanName, otherLoan, staffNo, adminStaffNo, loanApplyDate, loanReviewDate, applicantBasicSalary, applicantNetSalary

This function is to fetch some required data to be displayed in loan form.

```
$sql = "SELECT
            loan.loanID,
            loan.loanAmount,
            loan.loanDuration,
            loan.monthlyPayment,
            loan.bankName,
            loan.accountBankNumber,
            loanType.loanName,
            loan.otherLoan,
            loan.staffNo,
            loan.adminStaffNo,
            loan.loanApplyDate,
            loan.loanReviewDate,
            loan.applicantBasicSalary,
            loan.applicantNetSalary
        FROM loan
        LEFT JOIN loanType ON loan.loanType = loanType.loanType
        WHERE loan.staffNo = '$staffNo' AND loan.loanID = '$loanID'";
```

loanID	loanAmount	loanDuration	monthlyPayment	bankName	accountBankNumber	loanName	otherLoan
18	10000.00	24	434.17	Bank Simpanan Nasional (BSN)	123443211234	skim khas	

staffNo	adminStaffNo	loanApplyDate	loanReviewDate	applicantBasicSalary	applicantNetSalary
1110	NULL	2025-01-20	NULL	9310.00	9010.00

3. SELECT loanID, guarantorName, guarantorIC, guarantorID, guarantorPF, guarantorPhoneNumber

This function should retrieve some of the guarantor information to review in loan application form.

```
$sql = "
    SELECT
        loan.loanID,
        guarantorinfo.guarantorName,
        guarantorinfo.guarantorIC,
        guarantorinfo.guarantorID,
        guarantorinfo.guarantorPF,
        guarantorinfo.guarantorPhoneNumber
    FROM loan
    LEFT JOIN guarantorinfo ON loan.loanID = guarantorinfo.loanID
    WHERE loan.staffNo = ? AND loan.loanID = ?
";
```

loanID	guarantorName	guarantorIC	guarantorID	guarantorPF	guarantorPhoneNumber
18	BUNGA BINTI RASHID	700928099010	1009	9010	01110099010
18	FIRAZ BIN YUSOF	790112131456	6541	1456	01665411456

4. UPDATE loan status

Update the loan table with the provided adminStaffNo

```
$sql = "UPDATE loan
        SET loanStatus = $status,
            loanReviewDate = '$reviewDate',
            adminStaffNo = '$adminStaffNo'
        WHERE staffNo = '$staffNo' AND loanID = '$loanID'";
```

loanStatus	loanReviewDate
4	2025-01-21

5. SELECT email, firstName, lastName.

This function is to select email, first name and last name from the applicant table.

```
SELECT u.email, u.firstName, u.lastName
    FROM users u
    JOIN applicant a ON a.email = u.email
    WHERE a.staffNo = '1234';
```

email	firstName	lastName
nawwarahauni@graduate.utm.my	Nawwarah	Auni

6. INSERT INTO ApplicationFeedback table for loan application

```
INSERT INTO ApplicationFeedback (email, reason, dateGiven)
       VALUES (
           (SELECT u.email
            FROM users u
            JOIN applicant a ON a.email = u.email
            WHERE a.staffNo = '1234'),
           'SLIP GAJI TIDAK DISAHKAN',
           CURDATE()
       );
```

←T→	feedbackID	email	reason	dateGiven
Edit Copy Delete	26	nawwarahauni@graduate.utm.my	SLIP GAJI TIDAK DISAHKAN	2025-01-26

Review Terminate

1. SELECT applicant.staffNo, applicant.applicantName, membershipend.id, membershipend.applyDate, membershipend.status, membershipend.reviewDate, status.statusDesc

This select function is to select applicant information to be reviewed.

```
$sql = "SELECT
            applicant.staffNo,
            applicant.applicantName,
            membershipend.id,
            membershipend.applyDate,
            membershipend.status,
            membershipend.reviewDate,
            status.statusDesc
        FROM membershipend
        LEFT JOIN applicant ON membershipend.staffNo = applicant.staffNo
        LEFT JOIN status ON membershipend.status = status.status
        WHERE membershipend.status = 1";
```

staffNo	applicantName	id	applyDate	status	reviewDate	statusDesc
1234	NAWWARAH AUNI BINTI NAZRUDIN	1	2025-01-27	1	2025-01-27	disemak

2. UPDATE membershipend

This function is to update the status of membership termination applications after being reviewed.

```
$sql = "UPDATE membershipend
        SET status = $status,
            reviewDate = '$reviewDate',
            adminStaffNo = '$adminStaffNo'
        WHERE staffNo = '$staffNo'";
```

id	applyDate	endDate	reason	reviewDate	approveDate	staffNo	adminStaffNo	alkStaffNo	status
1	2025-01-27	NULL	Pencen	2025-01-27	NULL	1234	6543	NULL	5

3. SELECT staffNo, applicantName, applicantIC, applicantPF, applicantStreet, applicantPostcode, applicantCity, applicantState, applicantGender, applicantReligion, applicantRace, applicantPosition, officeStreet, officePostcode, applicantPhoneNumber, applicantDOB, applicantAge, officeFax, officeCity, applicantPhoneHome, applicantSalary, applicantPosition

This function is to select applicant information to display at admin's interface

```
$sql = "SELECT staffNo, applicantName, applicantIC, applicantPF, applicantStreet, applicantPostcode, applicantCity,
    applicantState, applicantGender, applicantReligion, applicantRace, applicantPosition, officeStreet, officePostcode,
    applicantPhoneNumber, applicantDOB, applicantAge, officeFax, officeCity, applicantPhoneHome, applicantSalary,
    applicantPosition
        FROM applicant
        WHERE staffNo = ?";
```

staffNo	applicantName	applicantIC	applicantPF	applicantStreet	applicantPostcode	applicantCity	applicantState	applicantGender	applicantReligion
1234	NAWWARAH AUNI BINTI NAZRUDIN	940518160909	5468	Jalan Lorong Berkat 1	47100	Puchong	Selangor	PEREMPUAN	ISLAM

applicantRace	applicantPosition	officeStreet	officePostcode	applicantPhoneNumber	applicantDOB
CINA	STAFF	JALAN KEMUBU 8/9	40320	01117371501	2004-05-18

applicantAge	officeFax	officeCity	applicantPhoneHome	applicantSalary	applicantPosition
25	0987654333	KOTA BHARU	0304121523	13000.00	STAFF

4. SELECT membership.reason

This function is to select reason for terminate applicant's membership.

```
$sql = "SELECT membershipend.reason
        FROM membershipend
        WHERE staffNo = ?";
```

reason
Pencen

4.2.2.5. Approval Module DML

APPROVE MEMBERSHIP

1. INSERT INTO ALK

```
INSERT INTO `alk` (`staffNo`, `email`) VALUES
(9991, 'aminahkassim@gmail.com');
```

	staffNo	email
Edit Copy Delete	9991	aminahkassim@gmail.com

2. UPDATE membershipStatus, membershipApproveDate and alkStaffNo into membership table. This happened after alk approve application

```
$sql = "UPDATE membership
        SET membershipStatus = $status,
            membershipApproveDate = " . ($approvalDate ? "'$approvalDate'" : "NULL") . ",
            alkStaffNo = " . ($alkStaffNo ? "'$alkStaffNo'" : "NULL") . "
        WHERE staffNo = '$staffNo'";
```

membershipStatus	membershipApproveDate	alkStaffNo	membershipApplyDate
2	2025-01-08	9991	2025-01-07
3	2025-01-08	9991	2025-01-07

3. SELECT information from applicant table to be display in applicant information

```sql
SELECT
    applicant.staffNo,
    applicant.applicantName,
    applicant.applicantIC,
    applicant.applicantPF,
    applicant.applicantStreet,
    applicant.applicantPostcode,
    applicant.applicantCity,
    applicant.applicantState,
    applicant.applicantGender,
    applicant.applicantReligion,
    applicant.applicantRace,
    applicant.applicantPosition,
    applicant.officeStreet,
    applicant.officePostcode,
    applicant.officeCity,
    applicant.applicantPhoneNumber,
    applicant.applicantDOB,
    applicant.applicantAge,
    applicant.maritalStatus,
    applicant.applicantGrade,
    applicant.officeFax,
    applicant.applicantPhoneHome,
    applicant.applicantSalary,
    familyinfo.familyName,
    familyinfo.relationship,
    familyinfo.familyIC,
    bankinfo.bankAccountName,
    bankinfo.accountBankNumber,
    savingtype.feeMasuk, savingtype.modahSyer,
    savingtype.modalYuran, savingtype.wangDepositAnggota, savingtype.sumbanganTabung,
    savingtype.simpananTetap
FROM applicant
LEFT JOIN familyinfo ON applicant.staffNo = familyinfo.staffNo
LEFT JOIN bankinfo ON applicant.staffNo = bankinfo.staffNo
LEFT JOIN savingtype ON applicant.staffNo = savingtype.staffNo
WHERE applicant.staffNo = '$staffNo'
```

staffNo	applicantName	applicantIC	applicantPF	applicantStreet	applicantPostcode	applicantCity	applicantState	applicantGender	applicantReligion	applicantRace
3331	FATIN SUHAIMI BINTI AHMAD	40518160765	1113	JALAN KEMUBU 8/9	40320	KOTA BHARU	KELANTAN	P	ISLAM	MELAYU
3331	FATIN SUHAIMI BINTI AHMAD	40518160765	1113	JALAN KEMUBU 8/9	40320	KOTA BHARU	KELANTAN	P	ISLAM	MELAYU

applicantPosition	officeStreet	officePostcode	officeCity	applicantPhoneNumber	applicantDOB	applicantAge	maritalStatus	applicantGrade	officeFax	applicantPhoneHome
STAF	JALAN KEMUBU 8/9	40320	KOTA BHARU	195108779	1980-07-15	44	BERKAHWIN	D	98765433	-
STAF	JALAN KEMUBU 8/9	40320	KOTA BHARU	195108779	1980-07-15	44	BERKAHWIN	D	98765433	-

applicantSalary	familyName	relationship	familyIC	bankAccountName	accountBankNumber	feeMasuk	modahSyer	modalYuran	wangDepositAnggota	sumbanganTabung
7000.00	LAILA BINTI HAKIM	ANAK-BERANAK	405160765	RHB	653274395	50	100	100.00	55.00	100.00
7000.00	ALIA BINTI HAKIM	ANAK-BERANAK	4051716543	RHB	653274395	50	100	100.00	55.00	100.00

simpananTetap
200.00
200.00

4. SELECT applicant's family information to be display in applicant's information page

```
SELECT familyinfo.familyName, familyinfo.relationship, familyinfo.familyIC
FROM familyinfo
WHERE familyinfo.staffNo = '$staffNo'
```

familyName	relationship	familyIC
LAILA BINTI HAKIM	ANAK-BERANAK	405160765
ALIA BINTI HAKIM	ANAK-BERANAK	4051716543

5. SELECT information from table applicant, status and membership to be display in membership list page

```
$sql = "SELECT
            applicant.staffNo,
            applicant.applicantName,
            membership.membershipApplyDate,
            membership.membershipStatus,
            membership.membershipApproveDate,
            membership.alkStaffNo,
            status.statusDesc
        FROM membership
        LEFT JOIN applicant ON membership.staffNo = applicant.staffNo
        LEFT JOIN status ON membership.membershipStatus = status.status
        WHERE membership.membershipStatus=5
```

staffNo	applicantName	membershipApplyDate	membershipStatus	membershipApproveDate	alkStaffNo	statusDesc
3331	FATIN SUHAIMI BINTI AHMAD	2025-01-07	5	2025-01-08	9991	Lengkap

APPROVE LOAN

1. SELECT attributes from applicant, loan and status table to be display in loan list page

```
$sql = "SELECT
            applicant.staffNo,
            applicant.applicantName,
            loan.loanID,
            loan.loanApplyDate,
            loan.loanStatus,
            loan.loanApproveDate,
            loan.alkStaffNo,
            loan.loanID,
            status.statusDesc
        FROM loan
        LEFT JOIN applicant ON loan.staffNo = applicant.staffNo
        LEFT JOIN status ON loan.loanStatus = status.status
        WHERE loan.loanStatus =5
        ";
```

staffNo	applicantName	loanID	loanApplyDate	loanStatus	loanApproveDate	alkStaffNo	loanID	statusDesc
6661	SAFIYA ALI BINTI AIMAN	10	2025-01-14	5	2025-01-27	NULL	10	Lengkap

2. UPDATE loanStatus, lonaApproveDate and alkStaffNo in loan table after ALK approve or reject application

```
$sql = "UPDATE loan
        SET loanStatus = $status,
            loanApproveDate = $approvalDateValue,
            alkStaffNo = $alkStaffNoValue
        WHERE staffNo = '$staffNo' AND loanID = '$loanID'";
```

loanStatus	loanApproveDate	alkStaffNo
2	2025-01-15	9991

3. SELECT attributes from applicant and loan table to be display in applicant loan information page

```
$sql = "
    SELECT
        applicant.applicantName,
        applicant.applicantIC,
        applicant.applicantDOB,
        applicant.applicantAge,
        applicant.applicantStreet,
        applicant.applicantPostcode,
        applicant.applicantGender,
        applicant.applicantReligion,
        applicant.applicantRace,
        applicant.applicantPF,
        applicant.applicantPosition,
        applicant.officeStreet,
        applicant.officePostcode,
        applicant.applicantPhoneNumber,
        loan.loanID
    FROM applicant
    LEFT JOIN loan ON loan.staffNo = applicant.staffNo
    WHERE applicant.staffNo = '$staffNo'  AND loan.loanID = '$loanID'";
```

applicantName	applicantIC	applicantDOB	applicantAge	applicantStreet	applicantPostcode	applicantGender	applicantReligion	applicantRace	applicantPF	applicantPosition
FATIN SUHAIMI BINTI AHMAD	40518160765	1980-07-15	44	JALAN KEMUBU 8/9	40320	P	ISLAM	MELAYU	1113	STAF

officeStreet	officePostcode	applicantPhoneNumber	loanID
JALAN KEMUBU 8/9	40320	195108779	1

4. SELECT attributes from loan and loanType table to be display in loan information table

```
$sql = "
    SELECT
        loan.loanType,
        loan.otherLoan,
        loan.loanAmount,
        loan.loanDuration,
        loan.monthlyPayment,
        loan.applicantNetSalary,
        loan.applicantBasicSalary,
        loan.loanID,
        loan.loanApplyDate,
        loan.loanReviewDate,
        loan.pengesahanMajikan,
        loantype.loanName
    FROM loan
    JOIN loantype ON loan.loanType = loantype.loanType
    WHERE loan.staffNo = '$staffNo' AND loan.loanID = '$loanID'
";
```

loanType	otherLoan	loanAmount	loanDuration	monthlyPayment	applicantNetSalary	applicantBasicSalary	loanID	loanApplyDate	loanReviewDate	pengesahanMajikan	loanName
2	-	5000.00	2	123.50	7000.00	5500.00	1	2024-12-09	NULL		al-innah

5. SELECT attributes from loan table to be display in applicant's bank information page

```
SELECT
    loan.loanID,
    loan.bankName,
    loan.accountBankNumber
FROM loan
WHERE loan.staffNo = '$staffNo' AND loan.loanID = '$loanID'
```

loanID	bankName	accountBankNumber
1	CIMB	7640568245

6. SELECT attributes from guarantorInfo table to display guarantor information in applicant's guarantor information page

```
SELECT
    guarantorIC,
    loanID,
    guarantorName,
    guarantorID,
    guarantorPF,
    guarantorPhoneNumber,
    photo
FROM guarantorInfo
WHERE loanID = '$loanID'
```

guarantorIC	loanID	guarantorName	guarantorID	guarantorPF	guarantorPhoneNumber	photo
040130130000	1	FARA BINTI ALIF	1234	4567	0118976543	
072319164330	1	HANI ZULAIKHA BINTI LUKMAN	5446	7611	0164432865	

7. SELECT attributes from loan and applicant table to be display in compare page

```
$sql = "SELECT
            loan.loanID,
            loan.loanAmount,
            loan.loanDuration,
            loan.monthlyPayment,
            loan.staffNo,
            loan.applicantNetSalary AS applicantNetSalary
        FROM loan
        LEFT JOIN applicant ON loan.staffNo = applicant.staffNo
        WHERE loan.staffNo = '$staffNo'";
```

loanID	loanAmount	loanDuration	monthlyPayment	staffNo	applicantNetSalary
1	5000.00	2	123.50	3331	7000.00
4	6000.00	2	123.50	3331	7000.00

8. SELECT attributes from applicant, membership, historymodimem and status table to be display in Sejarah Permohonan Anggota page

```sql
SELECT
            applicant.staffNo,
            applicant.applicantName,
            membership.membershipID,
            membership.membershipApplyDate,
            membership.membershipStatus,
            membership.membershipApproveDate,
            membership.alkStaffNo,
            historymodimem.newMemStatus,
            historymodimem.memChangeDate,
            historymodimem.memModifiedBy,
            status.statusDesc AS currentStatusDesc,
            newStatusDesc.statusDesc AS newStatusDesc
        FROM membership
        LEFT JOIN applicant ON membership.staffNo = applicant.staffNo
  LEFT JOIN status ON membership.membershipStatus = status.status
  LEFT JOIN historymodimem ON membership.membershipID = historymodimem.membershipID
  LEFT JOIN status AS newStatusDesc ON historymodimem.newMemStatus = newStatusDesc.status
  WHERE membership.membershipStatus IN (2, 3)
```

staffNo	applicantName	membershipID	membershipApplyDate	membershipStatus	membershipApproveDate	alkStaffNo	newMemStatus	memChangeDate	memModifiedBy
7771	SUFIAN SUHAIMI BIN AMRI	2	2025-01-07	3	2025-01-08	9991	3	2025-01-27	9991
1111	DAYANG FARAH BINTI ABANG IDHAM	3	2025-01-01	3	2025-01-09	9991	3	2025-01-27	9991
6661	SAFIYA ALI BINTI AIMAN	4	2025-01-01	2	2025-01-27	9991	NULL	NULL	NULL
9999	FARIS BIN FIRAS	5	2025-01-01	2	2025-01-27	9991	NULL	NULL	NULL

9. UPDATE newMemStatus, memChangeDate and modifiedBy in historymodimem table after modification happened

```sql
UPDATE historymodimem
SET newMemStatus = '$newStatus',
    memChangeDate = '$changeDate',
    memModifiedBy = '$alkStaffNo'
WHERE membershipID = '$membershipID'
```

newMemStatus New status after modification	memChangeDate Date status modified	memModifiedBy Modifier's staff number	membershipID
2	2025-01-27	9991	1
3	2025-01-27	9991	2
3	2025-01-27	9991	3

10. UPDATE newLoanStatus, loanChangeDate and loanModifiedBy in historymodiloan table after modification happened in loan application

```
SELECT
            applicant.staffNo,
            applicant.applicantName,
            loan.loanID,
            loan.loanApplyDate,
            loan.loanStatus,
            loan.loanApproveDate,
            loan.alkStaffNo,
            historymodiloan.newLoanStatus,
            historymodiloan.loanChangeDate,
            historymodiloan.loanModifiedBy,
            status.statusDesc AS currentStatusDesc,
            newStatusDesc.statusDesc AS newStatusDesc
        FROM loan
        LEFT JOIN applicant ON loan.staffNo = applicant.staffNo
    LEFT JOIN status ON loan.loanStatus = status.status
    LEFT JOIN historymodiloan ON loan.loanID = historymodiloan.loanID
    LEFT JOIN status AS newStatusDesc ON historymodiloan.newLoanStatus = newStatusDesc.status
    WHERE loan.loanStatus IN (2, 3)
```

staffNo	applicantName	loanID	loanApplyDate	loanStatus	loanApproveDate	alkStaffNo	newLoanStatus	loanChangeDate	loanModifiedBy	currentStatusDesc	newStatusDesc
3331	FATIN SUHAIMI BINTI AHMAD	1	2024-12-09	2	2025-01-15	9991	2	2025-01-27	9991	lulus	lulus
2222	NUR FARAH BINTI YUSOF	2	2025-01-10	3	NULL	NULL	3	2025-01-27	9991	ditolak	ditolak
7771	SUFIAN SUHAIMI BIN AMRI	3	2025-01-10	2	NULL	NULL	NULL	NULL	NULL	lulus	NULL
3331	FATIN SUHAIMI BINTI AHMAD	4	2025-01-08	2	NULL	NULL	NULL	NULL	NULL	lulus	NULL
6661	SAFIYA ALI BINTI AIMAN	10	2025-01-14	3	NULL	NULL	NULL	NULL	NULL	ditolak	NULL

11. UPDATE membershipStatus in membership table after gain new status from modification

```
UPDATE membership
 SET membershipStatus = '$newStatus'
 WHERE membershipID = '$membershipID'";
```

membershipStatus
2
3
3

12. INSERT email, reason and dateGiven into alkapplicationfeedback table. This happened when alk reject and give reason of the rejection

```
INSERT INTO alkApplicationFeedback (email, reason, dateGiven)
VALUES (
    (SELECT u.email
     FROM users u
     JOIN applicant a ON a.email = u.email
     WHERE a.staffNo = '$staffNo'),
    '$reason',
    CURDATE()
)";
```

feedbackID	email	reason	dateGiven
1	safiyamasnoor@gmail.com	Tidak letak slip gaji yang betul	2025-01-27

APPROVE TERMINATE MEMBERSHIP

13. SELECT applicant.staffNo, applicant.applicantName, membershipend.id, membershipend.applyDate, membershipend.status, membershipend.approveDate, membershipend.alkStaffNo, status.statusDesc. This function is to select application that has status = 5.

```
$sql = "SELECT
        applicant.staffNo,
        applicant.applicantName,
        membershipend.id,
        membershipend.applyDate,
        membershipend.status,
        membershipend.approveDate,
        membershipend.alkStaffNo,
        status.statusDesc
    FROM membershipend
    LEFT JOIN applicant ON membershipend.staffNo = applicant.staffNo
    LEFT JOIN status ON membershipend.status = status.status
    WHERE membershipend.status = 5";
```

staffNo	applicantName	id	applyDate	status	approveDate	alkStaffNo	statusDesc
1234	NAWWARAH AUNI BINTI NAZRUDIN	1	2025-01-27	5	NULL	NULL	Lengkap

14. UPDATE membershipend. This function is to update membershipend status after being approved by ALK

```
$sql = "UPDATE membershipend
        SET status = $status,
            approveDate = $approvalDateValue,
            alkStaffNo = $alkStaffNoValue
        WHERE staffNo = '$staffNo'";
```

id	applyDate	endDate	reason	reviewDate	approveDate	staffNo	adminStaffNo	alkStaffNo	status
1	2025-01-27	2025-01-27	Pencen	2025-01-27	2025-01-27	1234	6543	9991	2

15. SELECT membershipend.id, membershipend.staffNo, applicant.applicantName, applicant.applicantPF

```
$sql = "SELECT
            membershipend.id,
            membershipend.staffNo,
            applicant.applicantName,
            applicant.applicantPF
        FROM membershipend
        LEFT JOIN applicant ON membershipend.staffNo = applicant.staffNo
        WHERE membershipend.staffNo = '$staffNo'";
```

id	staffNo	applicantName	applicantPF
1	1234	NAWWARAH AUNI BINTI NAZRUDIN	5468

16. SELECT staffNo, applicantName, applicantIC, applicantPF, applicantStreet, applicantPostcode, applicantCity, applicantState, applicantGender, applicantReligion, applicantRace, applicantPosition, officeStreet, officePostcode, applicantPhoneNumber, applicantDOB, applicantAge, officeFax, officeCity, applicantPhoneHome, applicantSalary, applicantPosition. This function is to select applicant information to display on ALK interface.

```
$sql = "SELECT staffNo, applicantName, applicantIC, applicantPF, applicantStreet, applicantPostcode, applicantCity,
    applicantState, applicantGender, applicantReligion, applicantRace, applicantPosition, officeStreet, officePostcode,
    applicantPhoneNumber, applicantDOB, applicantAge, officeFax, officeCity, applicantPhoneHome, applicantSalary,
    applicantPosition
        FROM applicant
        WHERE staffNo = ?";
```

staffNo	applicantName	applicantIC	applicantPF	applicantStreet	applicantPostcode	applicantCity	applicantState	applicantGender	applicantReligion	applicantRace
1234	NAWWARAH AUNI BINTI NAZRUDIN	940518160909	5468	Jalan Lorong Berkat 1	47100	Puchong	Selangor	PEREMPUAN	ISLAM	CINA

applicantPosition	officeStreet	officePostcode	applicantPhoneNumber	applicantDOB	applicantAge	officeFax	officeCity	applicantPhoneHome	applicantSalary	applicantPosition
STAFF	JALAN KEMUBU 8/9	40320	01117371501	2004-05-18	25	0987654333	KOTA BHARU	0304121523	13000.00	STAFF

17. SELECT membershipend.reason From membershipend WHERE staffNo = ?";

```
$sql = "SELECT membershipend.reason
        FROM membershipend
        WHERE staffNo = ?";
```

reason
Pencen

4.2.2.6. Financial Tracking Module

1. SELECT…FROM…LEFT JOIN…ON

Select attributes from applicant and savingType to be display in Transaksi Bulanan page.

```
$sql = "SELECT a.staffNo, a.applicantName, st.modahSyer, st.sumbanganTabung, st.simpananTetap
        FROM applicant a
        LEFT JOIN savingtype st ON a.staffNo = st.staffNo";
```

2. SELECT…FROM…INNER JOIN…LEFT JOIN…WHERE…GROUP BY

Select attributes from applicant, loan and loanrepayment to be display in Transaksi Bulanan page.

```
$sql = "SELECT a.staffNo, l.loanID, l.loanAmount, l.monthlyPayment,
               (l.loanAmount - IFNULL(SUM(lr.repaymentAmount), 0)) AS remainingLoan
        FROM applicant a
        INNER JOIN loan l ON a.staffNo = l.staffNo
        LEFT JOIN loanrepayment lr ON l.loanID = lr.loanID
        WHERE l.loanStatus = 2
        GROUP BY a.staffNo, l.loanID";
```

3. UPDATE…SET…WHERE

Update attributes in savingType to track the amount of each saving type.

```
$sql = "UPDATE savingtype SET
        modahSyer = modahSyer + $toModahSyer,
        simpananTetap = simpananTetap + $toSimpananTetap,
        sumbanganTabung = sumbanganTabung + 5
        WHERE staffNo = $staffNo";
```

4. INSERT INTO…VALUES

Insert values into saving table to track monthly transaction for saving.

```
$sql = "INSERT INTO saving (savingAmount, savingDate, savingDesc, savingType, staffNo)
        VALUES (55, NOW(), 'Potongan Gaji Bulanan', 'Potongan Gaji', $staffNo)";
```

5. SELECT…FROM…LEFT JOIN…WHERE…GROUP BY

Select attributes from loan and loanrepayment to find information about selected loanID.

```
$sql = "SELECT loanAmount, (loanAmount - IFNULL(SUM(repaymentAmount), 0)) AS remainingLoan
        FROM loan
        LEFT JOIN loanrepayment ON loan.loanID = loanrepayment.loanID
        WHERE loan.loanID = $loanID
        GROUP BY loan.loanID";
```

6. INSERT INTO…VALUES

Insert values into loanrepayment table to track monthly transaction table.

```
$sqlInsert = "INSERT INTO loanrepayment (loanID, repaymentAmount, repaymentDate, repaymentType, repaymentDesc, repaymentReceipt)
              VALUES ($loanID, $payment, NOW(), 'Potongan Gaji', '$description', $receiptValue)";
```

7. SELECT…FROM..ORDER BY

Select attributes from the applicant and display them in applicantName order on the Transaksi Tidak Tetap page.

```
$staff_query = "SELECT staffNo, applicantName FROM applicant ORDER BY applicantName";
```

8. SELECT…FROM…JOIN…ON

Select attribute from loan, loantype and applicant to display the loan that applicant have in Transaksi Tidak Tetap page.

```
$loans_query = "SELECT l.loanID, lt.loanName, a.applicantName, a.staffNo
                FROM loan l
                JOIN loantype lt ON l.loanType = lt.loanType
                JOIN applicant a ON l.staffNo = a.staffNo";
```

9. INSERT INTO…VALUES

Insert values into loanrepayement table to track irregular transactions for loan of the member.

```
$repayment_query = "INSERT INTO loanrepayment
                   (repaymentAmount, repaymentDate, repaymentDesc,
                    repaymentType, loanID, repaymentReceipt)
                   VALUES
                   ($amount, CURDATE(), '$description',
                    '$paymentType', $loanID, '$receiptNo')";
```

10. UPDATE…SET

Update the values in the savingType table to track irregular transactions for the savings of the member.

```
$update_query = "UPDATE savingtype SET modahSyer = modahSyer + $amount WHERE staffNo = $staffNo";
```

```
$update_query = "UPDATE savingtype SET modalYuran = modalYuran + $amount WHERE staffNo = $staffNo";
```

```
$update_query = "UPDATE savingtype SET wangDepositAnggota = wangDepositAnggota + $amount WHERE staffNo = $staffNo";
```

```
$update_query = "UPDATE savingtype SET sumbanganTabung = sumbanganTabung + $amount WHERE staffNo = $staffNo";
```

```
$update_query = "UPDATE savingtype SET simpananTetap = simpananTetap + $amount WHERE staffNo = $staffNo";
```

```
$update_query = "UPDATE savingtype SET lainLain = lainLain + $amount WHERE staffNo = $staffNo";
```

11. SELECT…FROM…LEFT JOIN…ON…WHERE…GROUP BY

Select attributes from the loan, loanRepayment, and loanType tables based on the applicant's staffNo login, and group by loanID to display loan information in applicant interface for Rekod Pembayaran Simpanan dan Pinjaman page.

```
$sqlLoans = "SELECT l.loanID, l.loanAmount, l.monthlyPayment, lt.loanName,
                    SUM(lr.repaymentAmount) AS totalRepaid,
                    (l.loanAmount - IFNULL(SUM(lr.repaymentAmount), 0)) AS remainingLoan
             FROM loan l
             LEFT JOIN loanrepayment lr ON l.loanID = lr.loanID
             LEFT JOIN loantype lt ON lt.loanType = l.loanType
             WHERE l.staffNo = '$staffNo'
             GROUP BY l.loanID";
```

12. SELECT…FROM…WHERE…ORDER BY…DESC

Select attributes loanRepayment table based on the applicant's staffNo login, and order by descending repayment date to display loan repayment transaction history in applicant interface for Rekod Pembayaran Simpanan dan Pinjaman page.

```
$sqlRepayments = "SELECT repaymentDate, repaymentAmount, repaymentDesc, repaymentReceipt
                  FROM loanrepayment
                  WHERE loanID = '$loanID'
                  ORDER BY repaymentDate DESC";
```

13. SELECT…FROM…WHERE…ORDER BY…DESC

Select attributes from saving table based on the applicant's staffNo login, and order by descending saving date to display saving transaction history in applicant interface for Rekod Pembayaran Simpanan dan Pinjaman page.

```
$sqlSavings = "SELECT savingID, savingAmount, savingDate, savingDesc, savingType, savingReceipt
               FROM saving
               WHERE staffNo = '$staffNo'
               ORDER BY savingDate DESC";
```

14. SELECT…FROM…WHERE

Select attributes from savingType table based on the applicant's staffNo login to display recent saving amount for each type in applicant interface for Rekod Pembayaran Simpanan dan Pinjaman page.

```
$sqlSavingTypes = "SELECT * FROM savingtype WHERE staffNo = '$staffNo'";
```

4.2.2.7. Statement Report Production Module

1. SELECT…FROM…JOIN…ON…LEFT JOIN…WHERE…ORDER BY…ASC

Select attributes from applicant, users, membership and managing to display list of approve members at Penyata Kewangan page.

```
$query = "SELECT a.email, a.staffNo, a.applicantName, u.email, m.membershipStatus, g.status, g.notificationLatestDate
            FROM applicant a
            JOIN users u ON a.email = u.email
            JOIN membership m ON a.staffNo = m.staffNo
            LEFT JOIN managing g ON a.email = g.email
            WHERE m.membershipStatus = 2
            ORDER BY a.staffNo ASC";
```

2. SELECT…FROM…LEFT JOIN…ON…WHERE

Select attributes from applicant, membership, loan and savingType to retrieve the information to generate financial statements.

```
$query = "SELECT
                a.staffNo, a.applicantName, a.applicantIC, a.applicantPF,
                s.feeMasuk, s.modahSyer, s.modalYuran, s.wangDepositAnggota, s.sumbanganTabung, s.simpananTetap,
                l.loanStatus, l.loanAmount, l.loanDuration, l.loanType
            FROM applicant a
            LEFT JOIN membership m ON a.staffNo = m.staffNo
            LEFT JOIN loan l ON a.staffNo = l.staffNo
            LEFT JOIN savingType s ON a.staffNo = s.staffNo
            WHERE a.staffNo = '$staffNo'";
```

3. INSERT INTO…VALUES

Insert values into managing table to track latest notification for financial statement.

```
$sql_insert = "INSERT INTO managing (status, notificationLatestDate, email)
                VALUES ('Berjaya dihantar!', '$currentDate', '$email')
                ON DUPLICATE KEY UPDATE
                status = 'Berjaya dihantar!',
                notificationLatestDate = '$currentDate'";
```

4. SELECT…FROM…JOIN…ON…WHERE…ORDER BY

Select attributes from applicant and membership table to display the list of approved and rejected membership applications to display on Penghantaran Status page.

```
$query_anggota = "SELECT a.staffNo, a.applicantName, a.email, m.membershipStatus
                  FROM applicant a
                  JOIN membership m ON a.staffNo = m.staffNo
                  WHERE m.membershipStatus IN (2, 3)
                  AND (m.sendStatus != 1 OR m.sendStatus IS NULL)
                  ORDER BY a.staffNo ASC";
```

5. SELECT…FROM…JOIN…ON…WHERE…ORDER BY

Select attributes from applicant and loan table to display the list of approved and rejected loan applications to display on Penghantaran Status page.

```
$query_pinjaman = "SELECT a.staffNo, a.applicantName, a.email, p.loanStatus, p.loanID
                   FROM applicant a
                   JOIN loan p ON a.staffNo = p.staffNo
                   WHERE p.loanStatus IN (2, 3)
                   AND (p.sendStatus != 1 OR p.sendStatus IS NULL)
                   ORDER BY a.staffNo ASC";
```

6. UPDATE…SET

Update the value to 1 for emails sent successfully and 2 for failed emails in the loan and membership tables.

```
$updateQuery = "UPDATE loan SET sendStatus = 1 WHERE loanID = ?";
```

```
"UPDATE loan SET sendStatus = 2 WHERE loanID = ?";
```

```
$updateQuery = "UPDATE membership SET sendStatus = 1 WHERE staffNo = ?";
```

```
$updateQuery = "UPDATE membership SET sendStatus = 2 WHERE staffNo = ?";
```

7. SELECT COUNT( ) …FROM…WHERE…AND

Select and calculate new member and loan applications for the current month at Laporan Koperasi page.

```
$queryNewMembersCurrent = "SELECT COUNT(*) as total FROM membership
                           WHERE MONTH(membershipApplyDate) = $currentMonth
                           AND YEAR(membershipApplyDate) = $currentYear";
```

```
$queryLoanApplicationsCurrent = "SELECT COUNT(*) as total FROM loan
                                 WHERE MONTH(loanApplyDate) = $currentMonth
                                 AND YEAR(loanApplyDate) = $currentYear";
```

8. SELECT SUM( )…FROM…WHERE..AND

Select and calculate total amount of approved loan applications for the current month to display at Laporan Koperasi page.

```
$queryApprovedLoansCurrent = "SELECT SUM(loanAmount) as total_amount FROM loan
                              WHERE MONTH(loanApproveDate) = $currentMonth
                              AND YEAR(loanApproveDate) = $currentYear
                              AND loanStatus = 2";
```

9. SELECT COUNT ( )...FROM…WHERE

Select and count total approved members of koperasi KADA to display at Laporan Koperasi page.

```
$queryTotalMembers = "SELECT COUNT(*) as total FROM membership WHERE membershipStatus = 2";
```

10. SELECT COUNT( )...FROM…WHERE

Select and count total approved members of koperasi KADA to display at Laporan Koperasi page.

```
$queryNewMembers = "SELECT COUNT(*) as total FROM membership
                    WHERE MONTH(membershipApplyDate) = $selectedMonth
                    AND YEAR(membershipApplyDate) = $selectedYear";
```

11. SELECT COUNT ( )...SUM ( )...FROM…WHERE

Select and count total, approved, rejected and pending member and loan applications for selected month and year to display at Laporan Koperasi page.

```
$queryMembership = "SELECT
    COUNT(*) as total_applications,
    SUM(CASE WHEN membershipStatus = 2 THEN 1 ELSE 0 END) as approved,
    SUM(CASE WHEN membershipStatus = 3 THEN 1 ELSE 0 END) as rejected,
    SUM(CASE WHEN membershipStatus IN (1,4,5) THEN 1 ELSE 0 END) as pending
FROM membership
WHERE MONTH(membershipApplyDate) = $selectedMonth
AND YEAR(membershipApplyDate) = $selectedYear";
```

```
$queryLoans = "SELECT
    COUNT(*) as total_applications,
    SUM(CASE WHEN loanStatus = 2 THEN 1 ELSE 0 END) as approved,
    SUM(CASE WHEN loanStatus = 3 THEN 1 ELSE 0 END) as rejected,
    SUM(CASE WHEN loanStatus IN (1,4,5) THEN 1 ELSE 0 END) as pending
FROM loan
WHERE MONTH(loanApplyDate) = $selectedMonth
AND YEAR(loanApplyDate) = $selectedYear";
```

12. SELECT COUNT( )...FROM…WHERE

Select and count total, approved and rejected member and loan applications and loan amount approved for a range of years to display at Laporan Koperasi page.

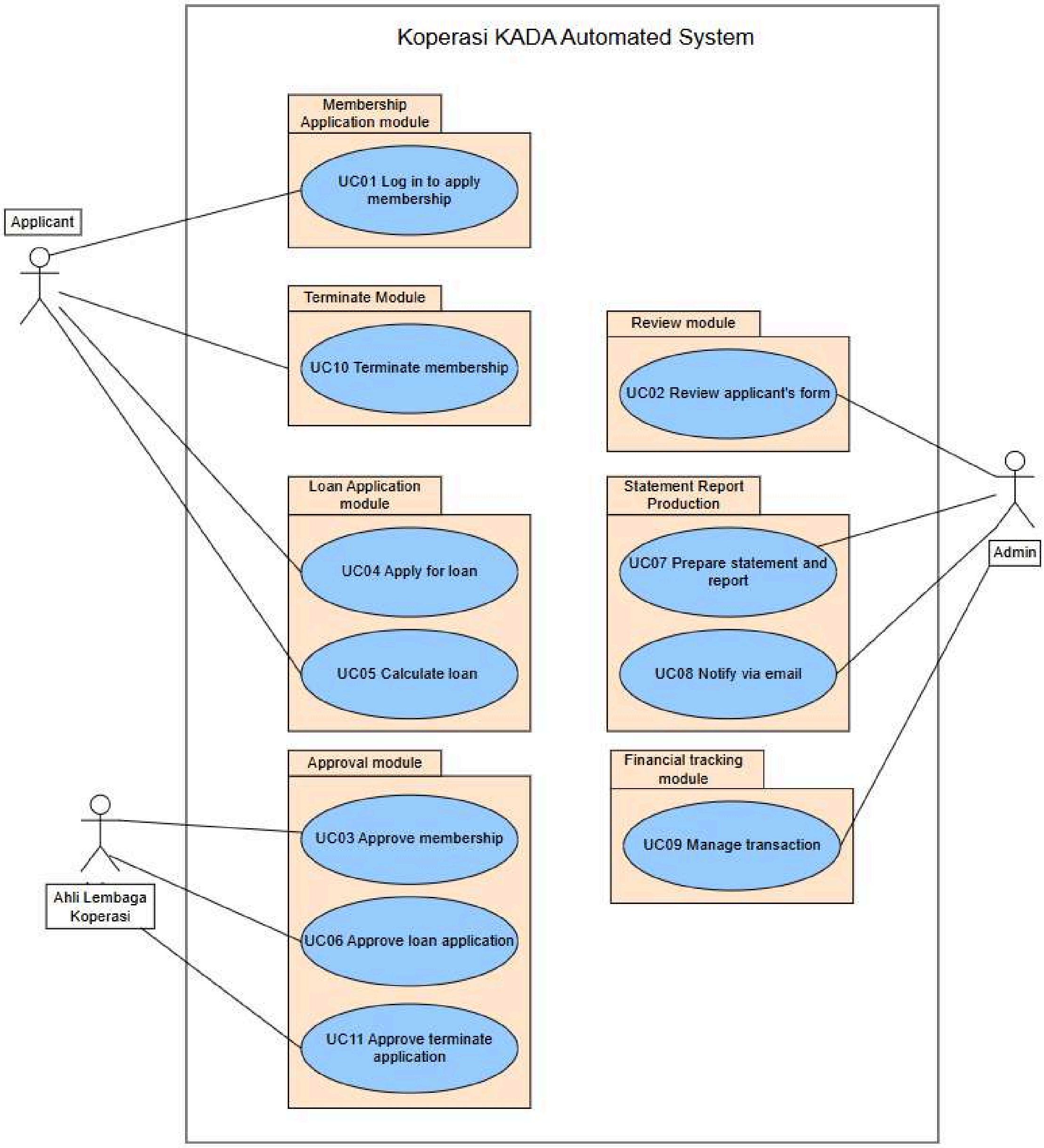
```
$queryAnnualGraph = "SELECT
    (SELECT COUNT(*) FROM membership WHERE membershipStatus = 2 AND YEAR(membershipApplyDate) <= $year) as total_members,
    (SELECT COUNT(*) FROM membership WHERE YEAR(membershipApplyDate) = $year) as total_member_applications,
    (SELECT COUNT(*) FROM membership WHERE membershipStatus = 2 AND YEAR(membershipApplyDate) = $year) as approved_members,
    (SELECT COUNT(*) FROM membership WHERE membershipStatus = 3 AND YEAR(membershipApplyDate) = $year) as rejected_members,
    (SELECT COUNT(*) FROM loan WHERE YEAR(loanApplyDate) = $year) as total_loan_applications,
    (SELECT COUNT(*) FROM loan WHERE loanStatus = 2 AND YEAR(loanApplyDate) = $year) as approved_loans,
    (SELECT COUNT(*) FROM loan WHERE loanStatus = 3 AND YEAR(loanApplyDate) = $year) as rejected_loans,
    (SELECT SUM(loanAmount) FROM loan WHERE loanStatus = 2 AND YEAR(loanApproveDate) = $year) as total_loan_amount";
```

5.0 Summary

To summarize, Koperasi KADA faces challenges with its inefficient and error prone paper-based system for managing membership and loan services. The whole process of doing things manually develops slow downs workflow and inaccuracies that could not meet the demand. In the proposed system, a digitalized Koperasi KADA system shall be introduced to introduce various submodules that will help smoothing the digital process of the system. The system automates membership registration, loan applications and their approvals, financial reporting while enabling tracking in real time and allowing data managed securely. This system enhances operational efficiency by reducing errors, improving transparency and minimizing paper-based records. In other words, it would smoothen the internal workflows and thereby enable Koperasi KADA to serve its members more effectively. All in all, digitalizing Koperasi KADA's system brings efficiency and an enhanced service to its members.

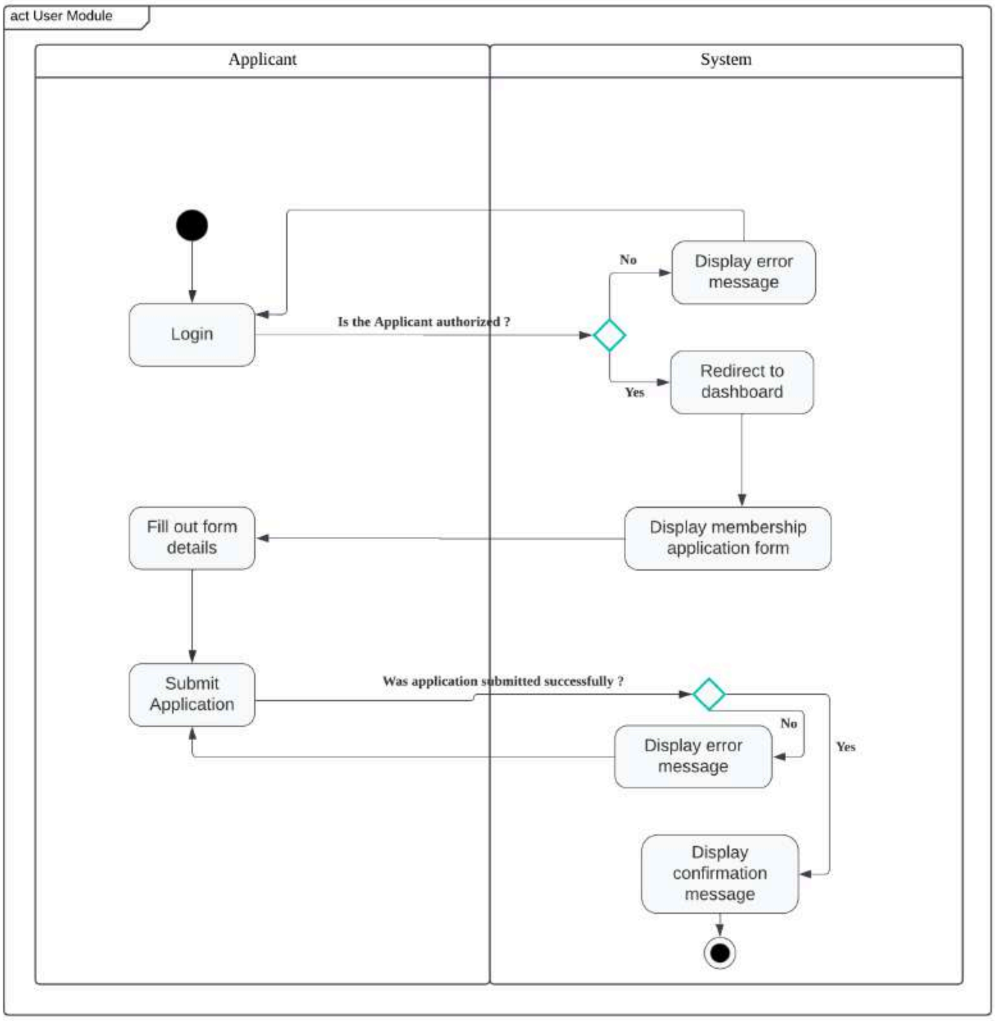
6.0 Appendices

6.1. Use Case Diagram

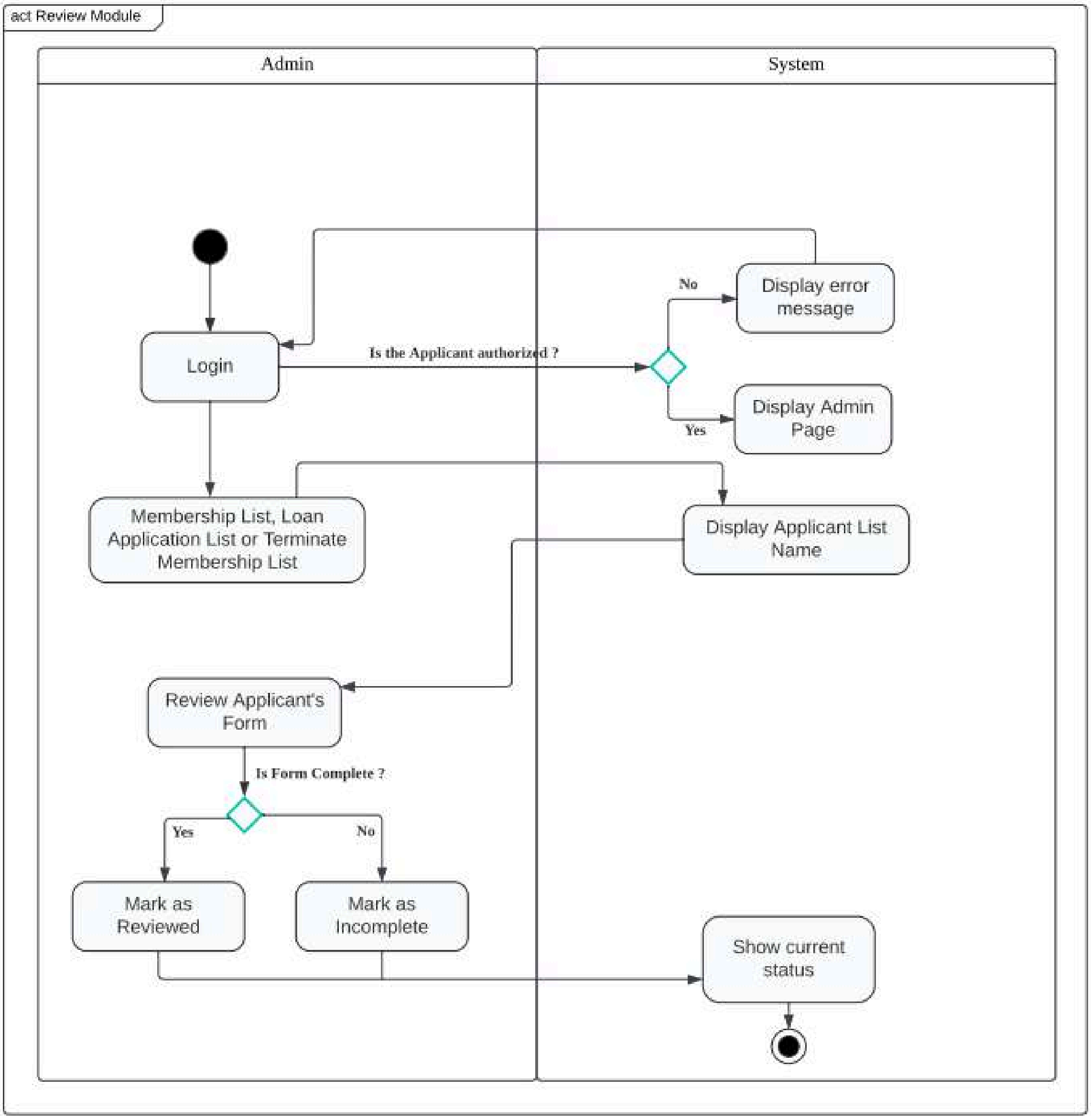


6.2. Activity Diagram

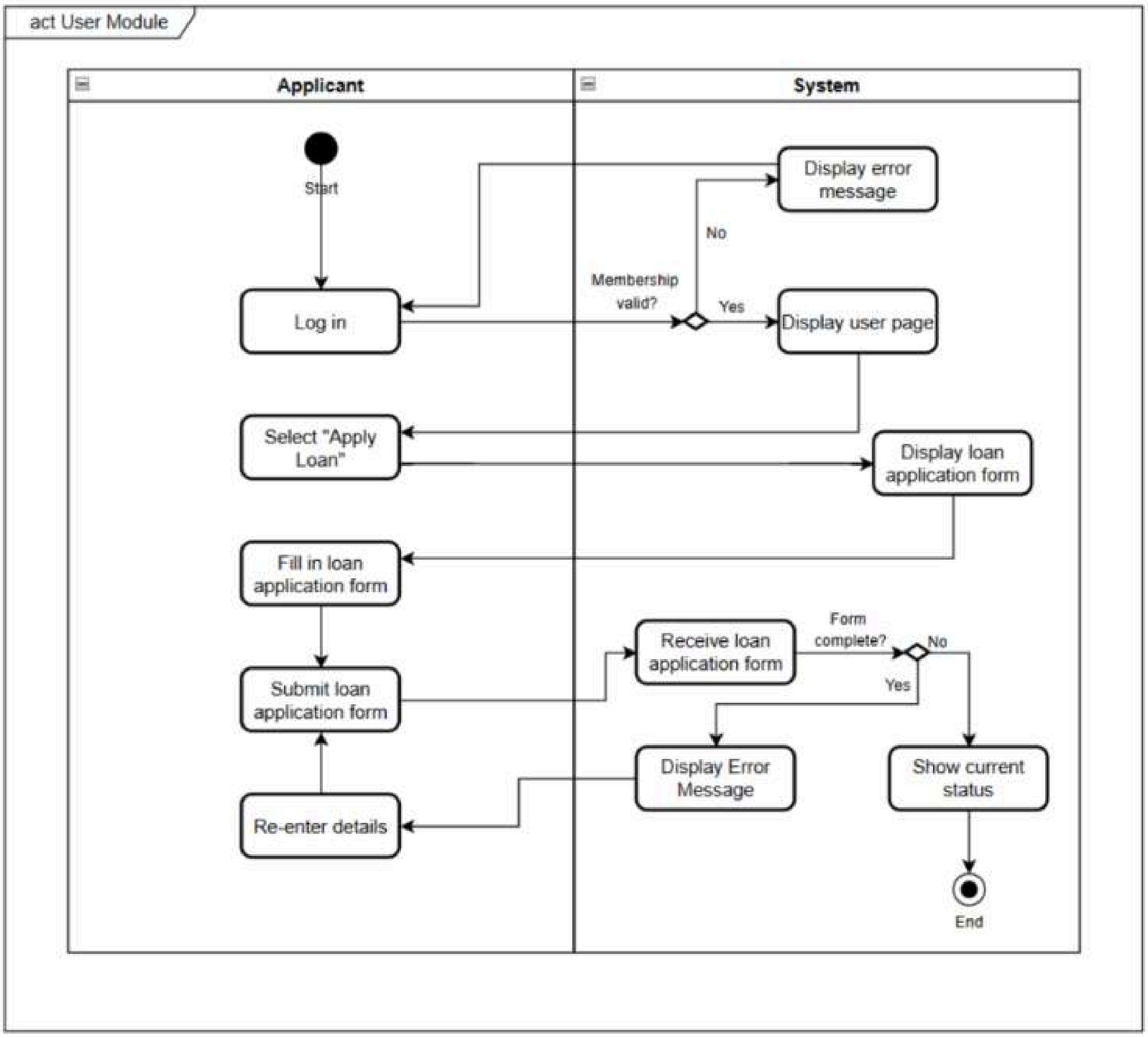
6.2.1 Activity Diagram Login to Apply Membership



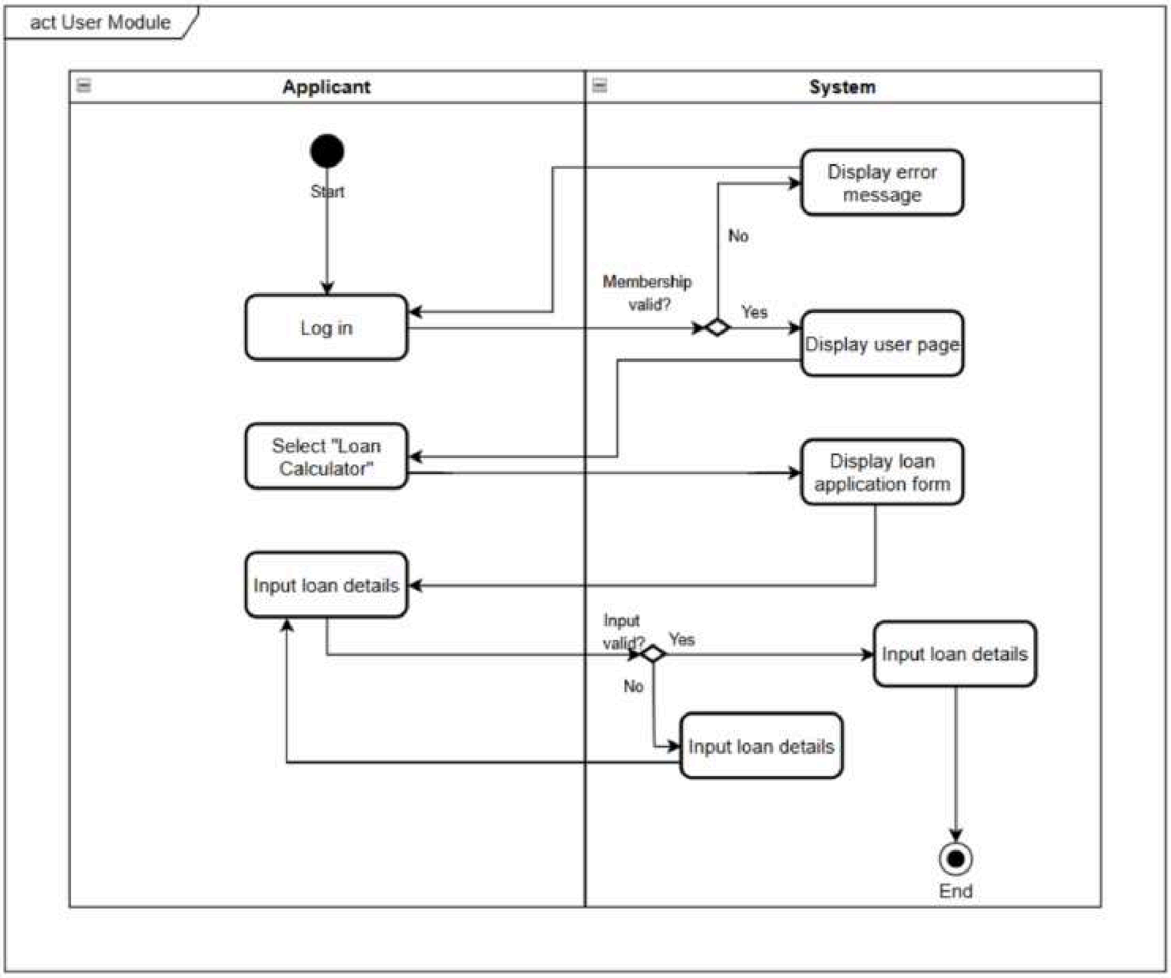
6.2.2 Activity Diagram Review Applicant's Form


act Review Module
Admin
System
Login
Is the Applicant authorized ?
No
Display error message
Yes
Display Admin Page
Membership List, Loan Application List or Terminate Membership List
Display Applicant List Name
Review Applicant's Form
Is Form Complete ?
Yes
No
Mark as Reviewed
Mark as Incomplete
Show current status

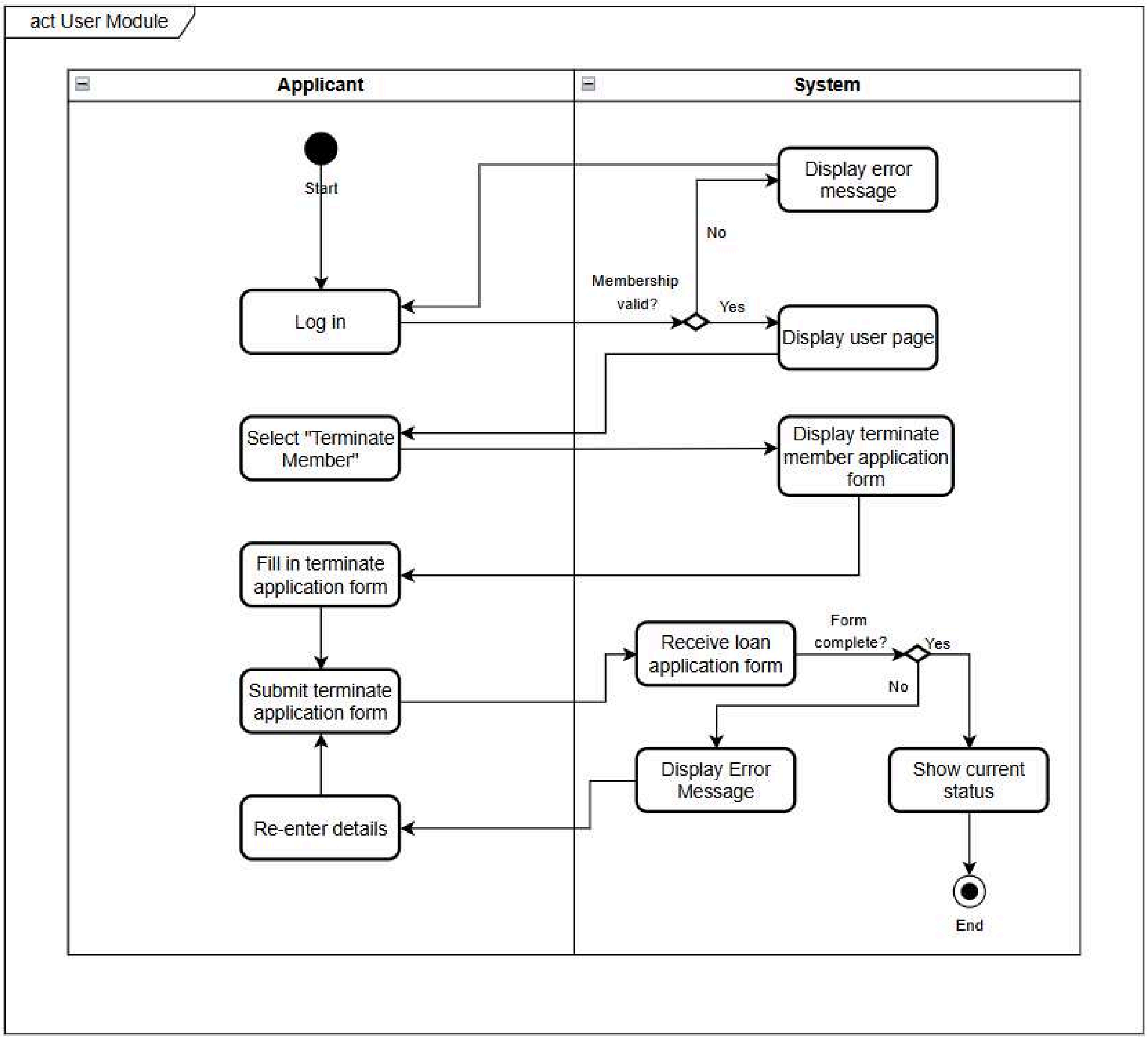
6.2.3 Activity Diagram for Apply Loan

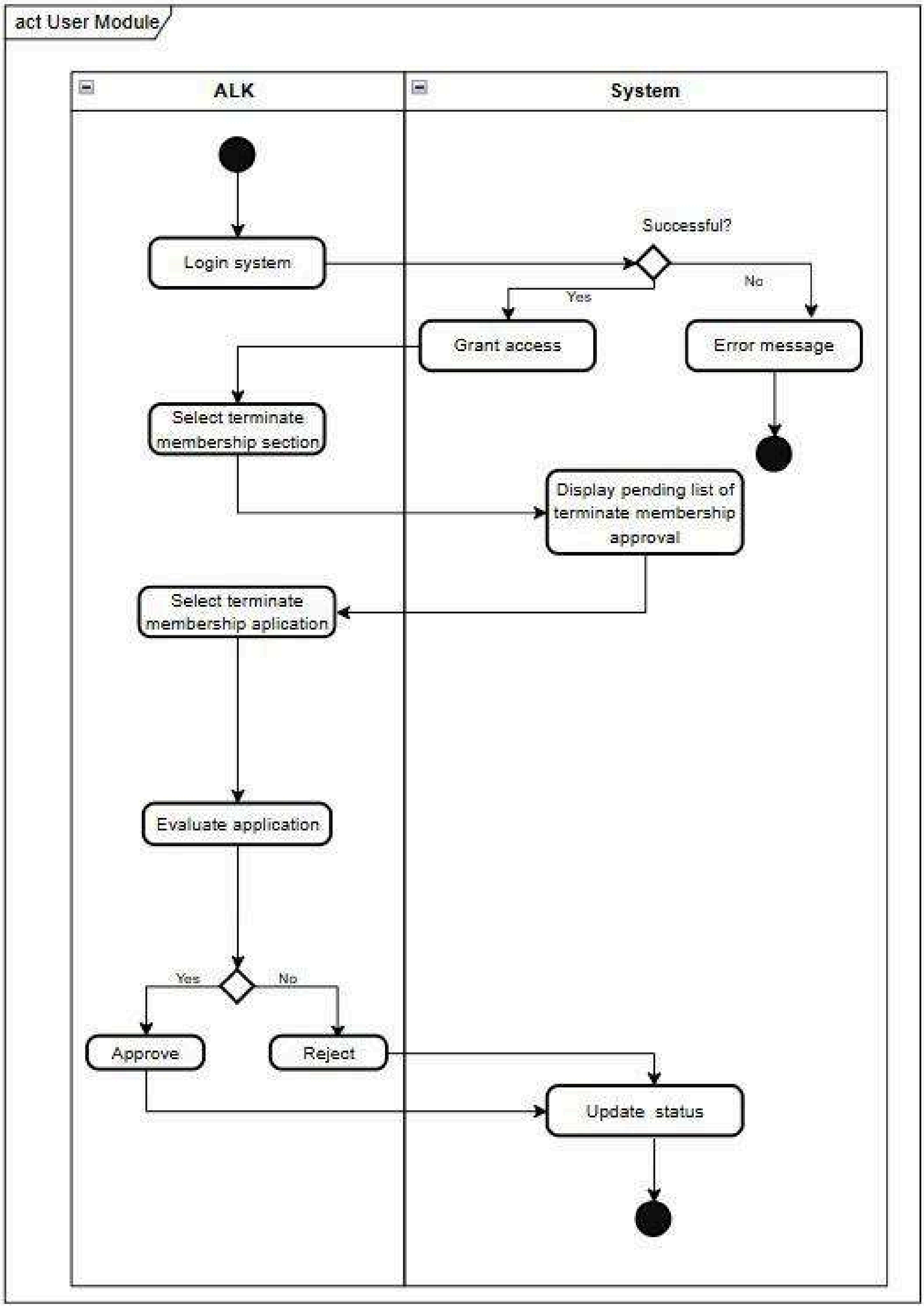


6.2.4 Activity Diagram for Calculate Loan

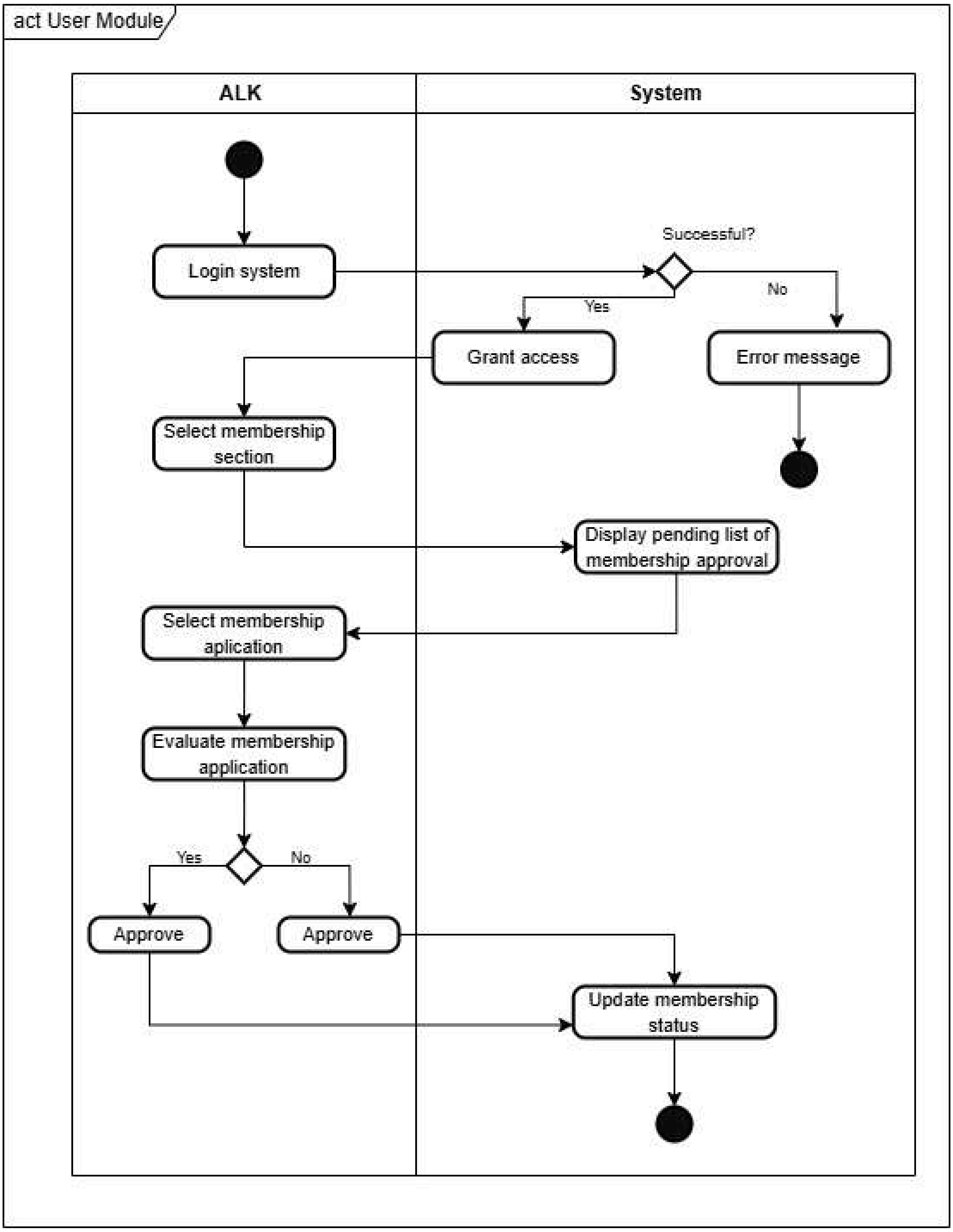


6.2.5 Activity Diagram for Terminate Membership

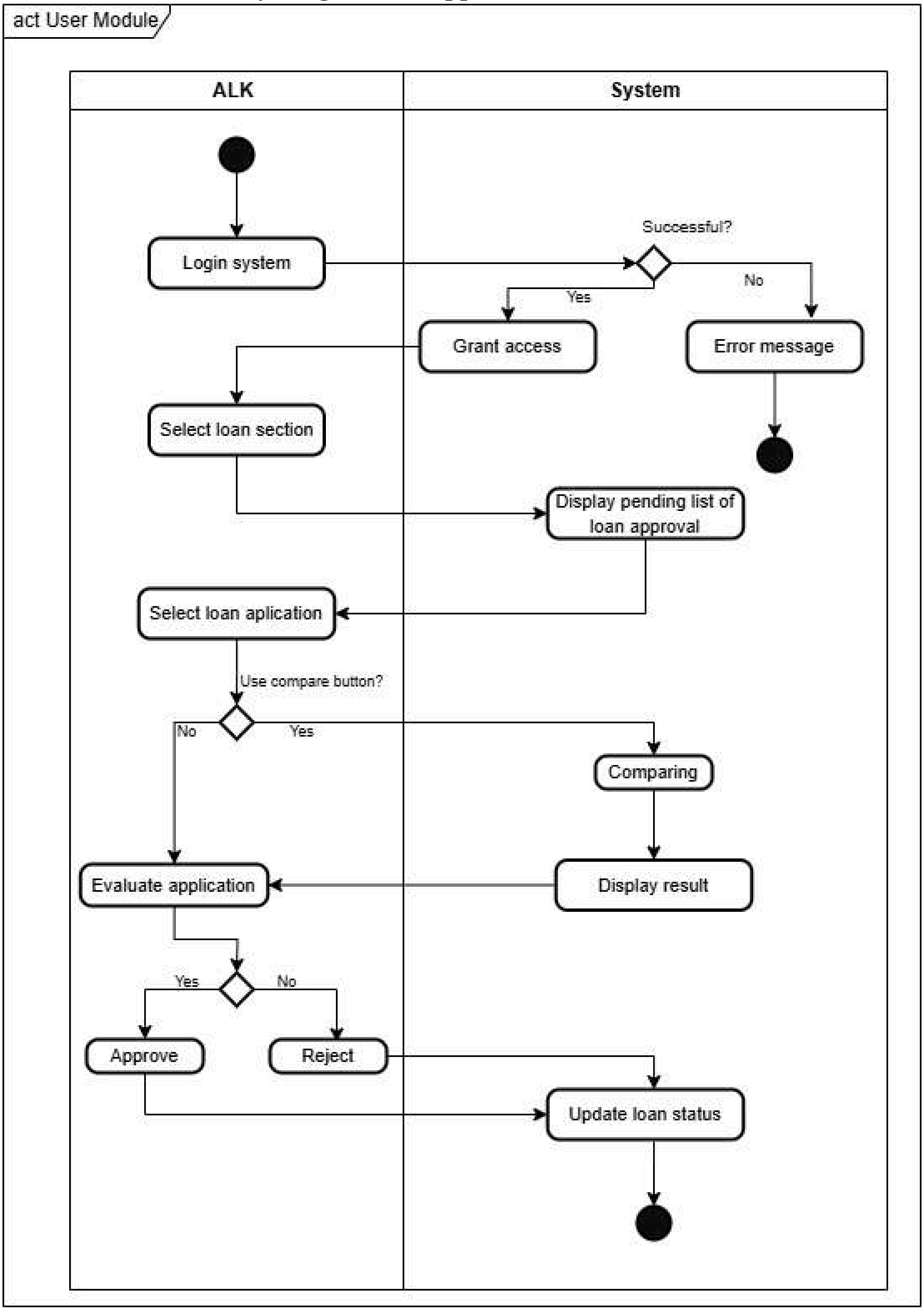

act User Module
Applicant
System
Start
Log in
Display error message
No
Membership valid?
Yes
Display user page
Select "Terminate Member"
Display terminate member application form
Fill in terminate application form
Submit terminate application form
Receive loan application form
Form complete?
Yes
No
Display Error Message
Show current status
Re-enter details
End



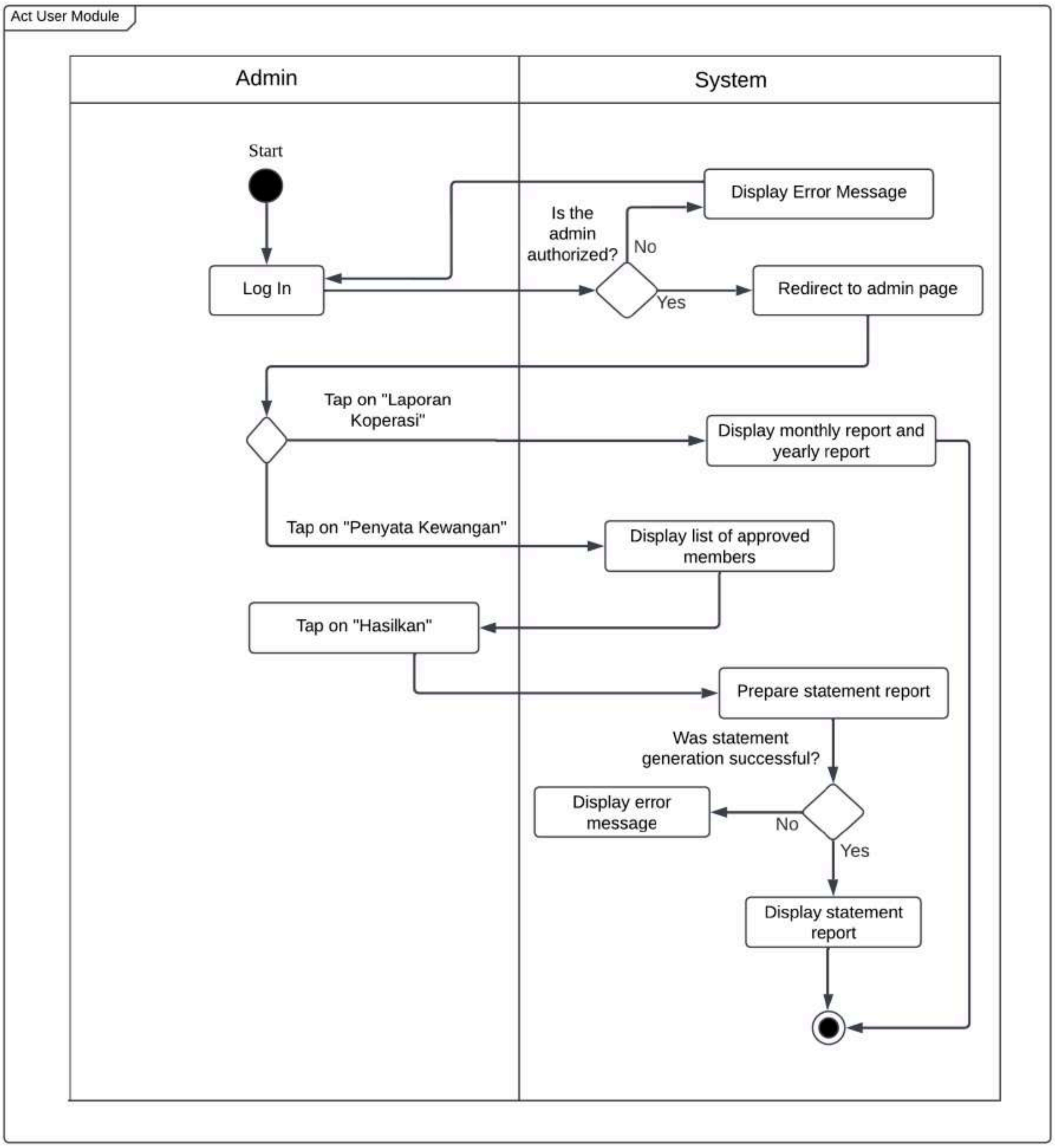
6.2.6 Activity Diagram for Approve Membership


act User Module
ALK
System
Login system
Successful?
Yes
No
Grant access
Error message
Select membership section
Display pending list of membership approval
Select membership aplication
Evaluate membership application
Yes
No
Approve
Approve
Update membership status

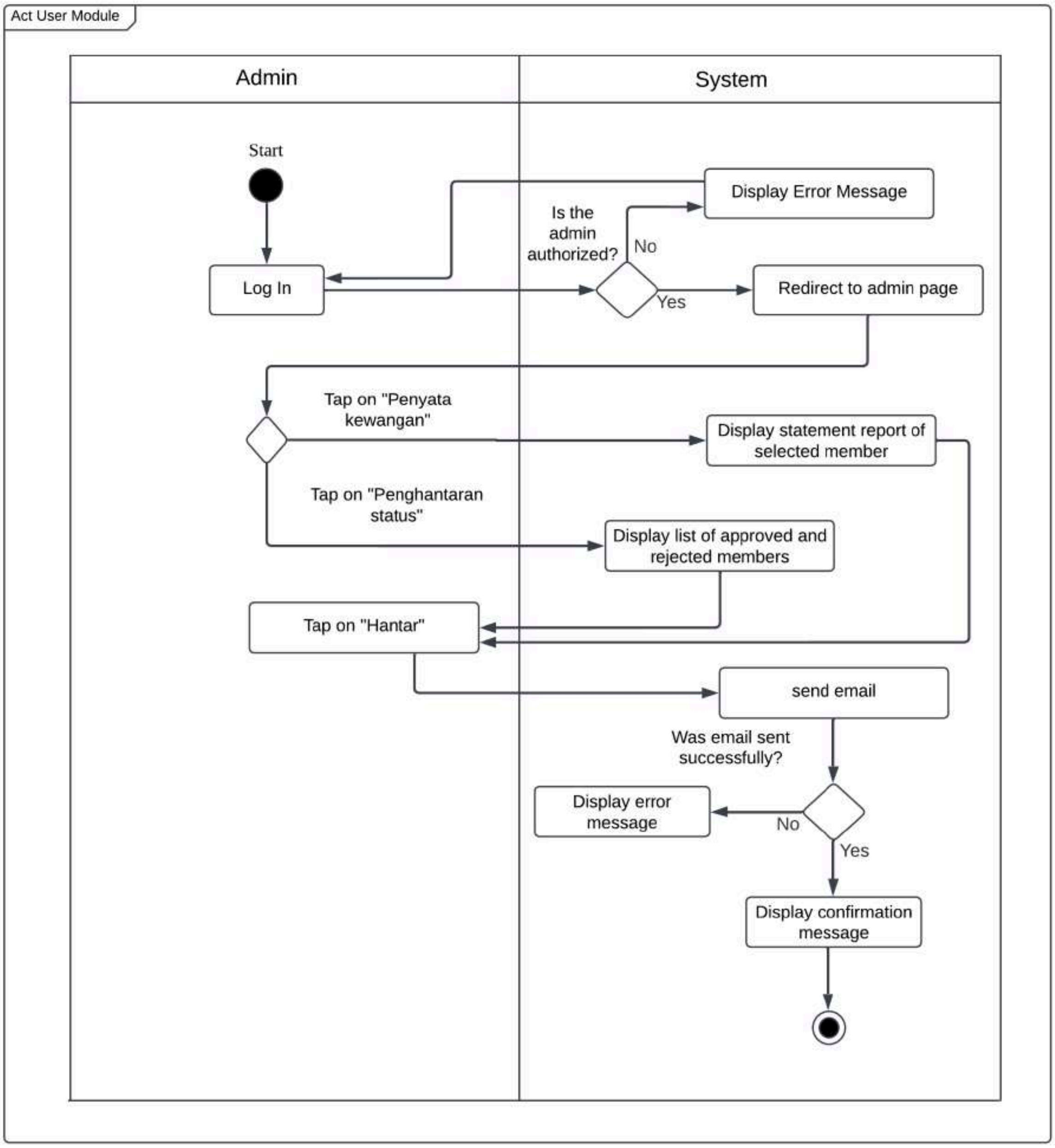
6.2.7 Activity Diagram for Approve Loan



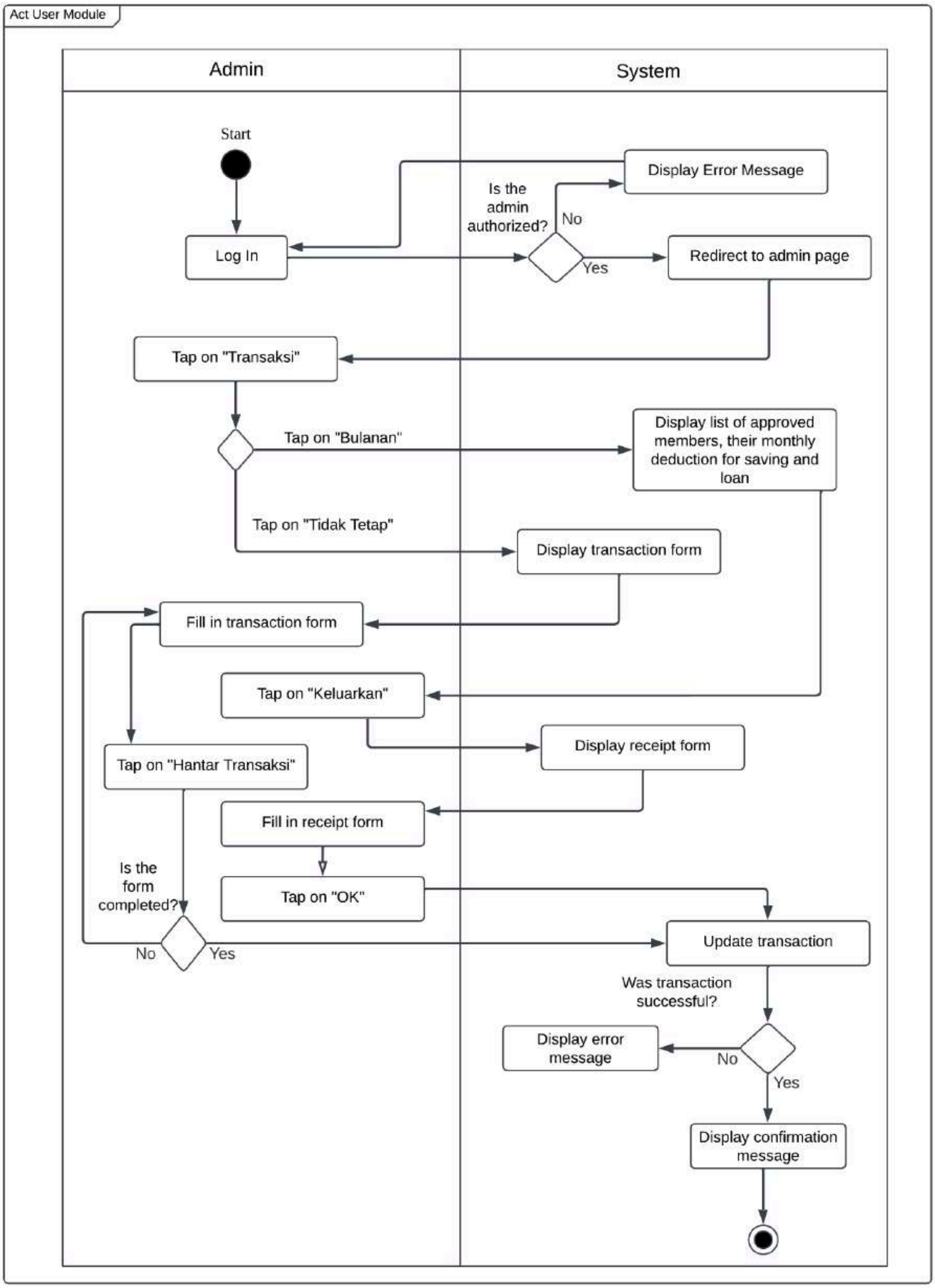
6.2.8 Activity Diagram for prepare statement and report


Act User Module
Admin
System
Start
Log In
Display Error Message
Is the admin authorized?
No
Yes
Redirect to admin page
Tap on "Laporan Koperasi"
Display monthly report and yearly report
Tap on "Penyata Kewangan"
Display list of approved members
Tap on "Hasilkan"
Prepare statement report
Was statement generation successful?
Display error message
No
Yes
Display statement report

6.2.9 Activity Diagram for Notify via Email



6.2.10 Activity Diagram for Manage Transaction



6.3. Meeting Log

DATE/TIME	25 January 2025 8:30 p.m.
LOCATION	Online (Whatsapp)
AGENDA	1. Discuss and review task in P3 2. Inform and add additional module 3. Task distribution between members
Meeting MC	Farra Nurzahin

ATTENDANCE

NAME	TIME	REASON FOR ABSENCE
DAYANG FARAH FARZANA BINTI ABANG IDHAM	2030	-
FARRA NURZAHIN BINTI ZAHARIL ANUAR	2030	-
NAWWARAH AUNI BINTI NAZRUDIN	2030	-
SAFIYA NURSYAHADAH BINTI MASNOOR	2030	-

MINUTES

No	Item Discussed	Ideas/Suggestion and Person Giving It	Person In Charge
1	Review tasks in P3	All members read and understand what needs to be done in P3.	All members
2	Additional module	Each member informs about the additional module and decides whether to add it or not.	All members
3	Task distribution	Farra distributed the tasks by spinning the wheel	All members